

INDUSTRIAL PROCESS MODELLING & SIMULATION TRAINING

2 – 4 August 2022

Part 2 : Flowsheeting, Separators and Heat Exchangers

About Myself

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Qualifications:

- PhD Chemical Engineering, Universiti Malaysia Pahang (UMP), 2020
- MSc. Chemical Engineering, Universiti Malaysia Pahang (UMP), 2015
- BSc. Chemical Engineerig, Universiti Malaysia Pahang (UMP), 2011
- Aspen Certified User in Aspen Plus (AspenTech)

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Topics

1.1 Constructing a Flowsheet

1.2 Separators

- Types of separators
- Solving Problems using Sensitivity Analysis

1.3 Heat Exchangers

- Heater model
- HeatX model
- Exchanger Design Rating

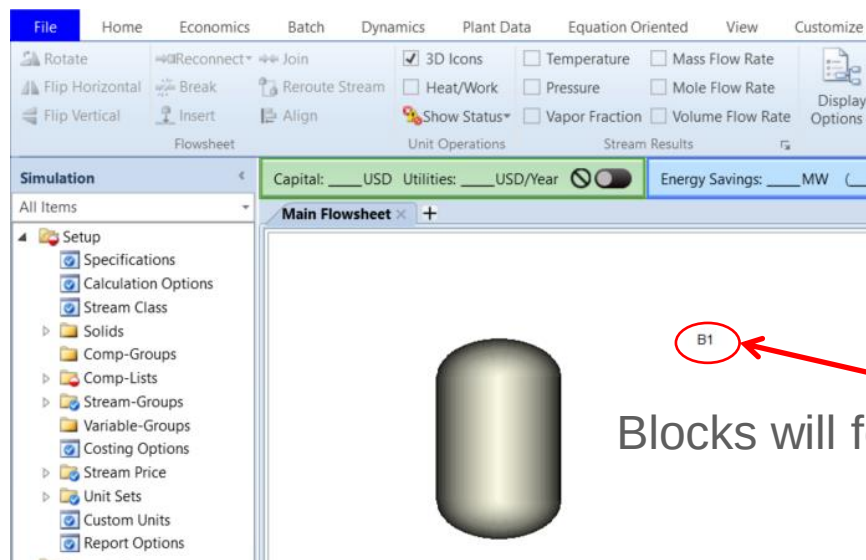
Outcomes

- Able to insert and connect streams and blocks in a flowsheet as well as reroute streams and insert blocks midstream and rotate and flip block icons
- Able to simulate separators and solve problems related to separation using sensitivity analysis
- Able to simulate simple and rigorous heat exchangers and solve problems related to heat exchangers

Constructing a Flowsheet

Adding blocks to flowsheet

- To add a block to your flowsheet select the correct category and unit from the model palette and drag and drop onto the main flowsheet
- For example to add a flash2 to the flowsheet look for flash2 under separators, click on it and click on the main flowsheet

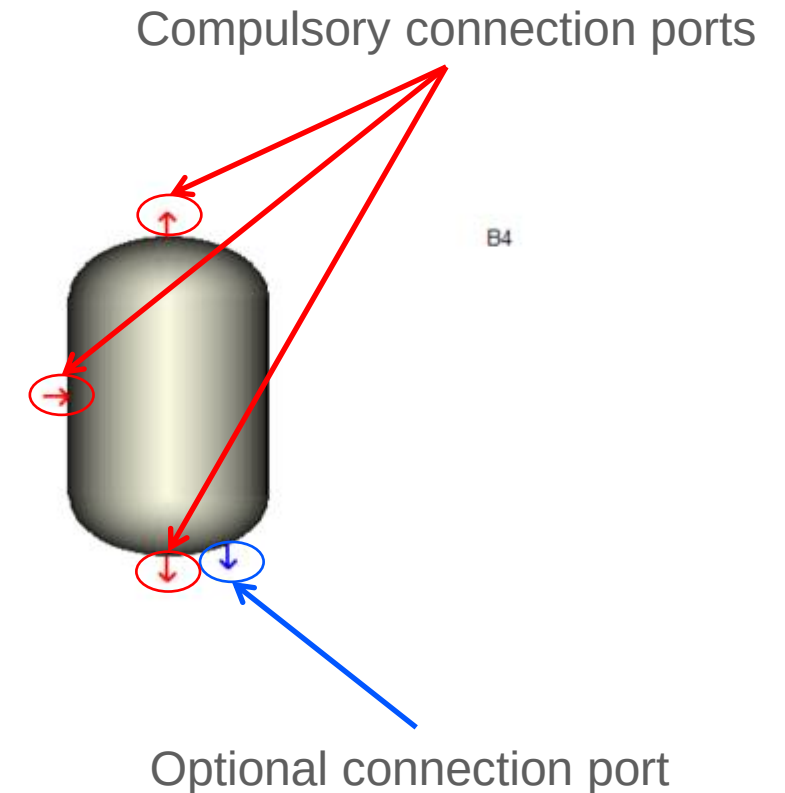


Flash 2 can be found under separators in the model palette

Blocks will follow this default naming convention, double click to rename

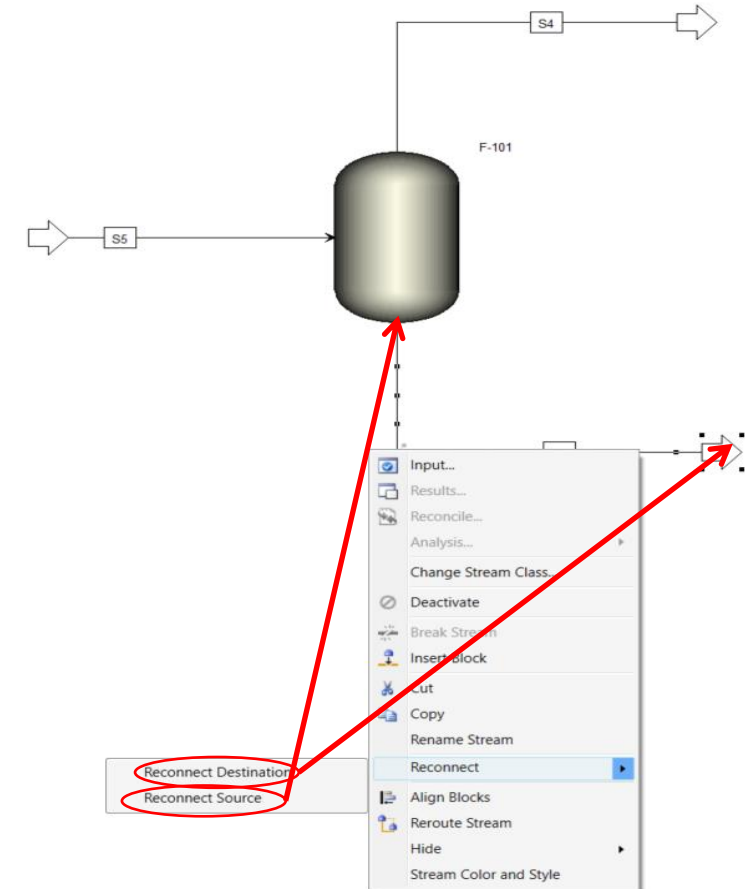
Connecting streams to blocks

- To connect streams to the unit, left click the material stream and as you move your cursor to the main flowsheet, the compulsory connections on the blocks present on the flowsheet will appear red indicating these need to be connected
- Simply left click on the compulsory connection port and left click again anywhere on the flowsheet
- Right click or hit the esc key to exit insert mode



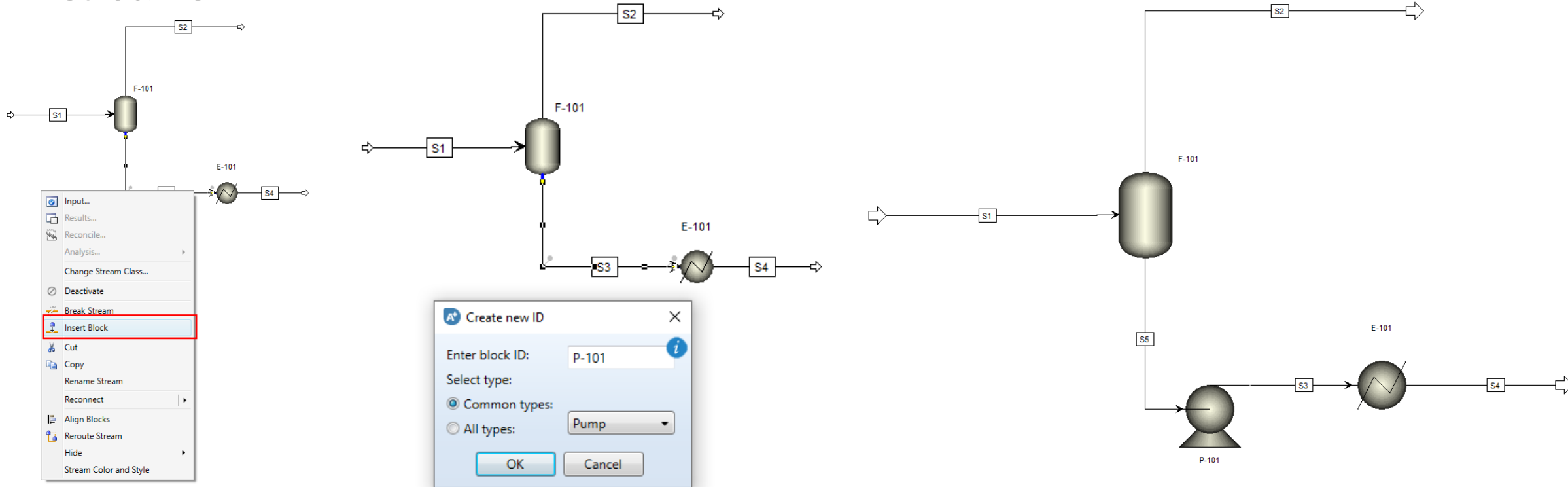
Deleting and reconnecting streams

- You can also delete streams or blocks by selecting them and hitting the Del key.
- To reconnect streams right click on a stream and select reconnect destination or source as desired



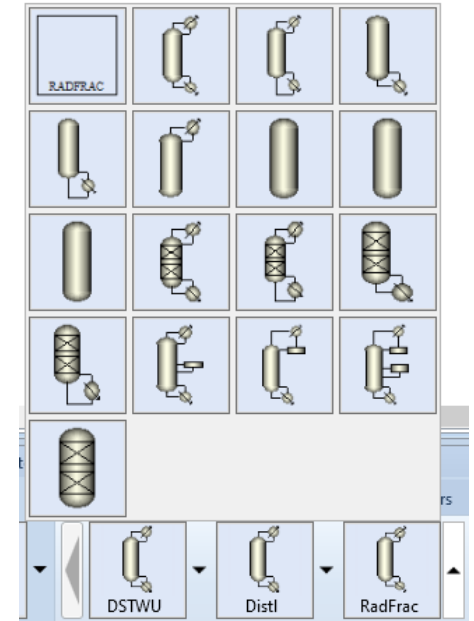
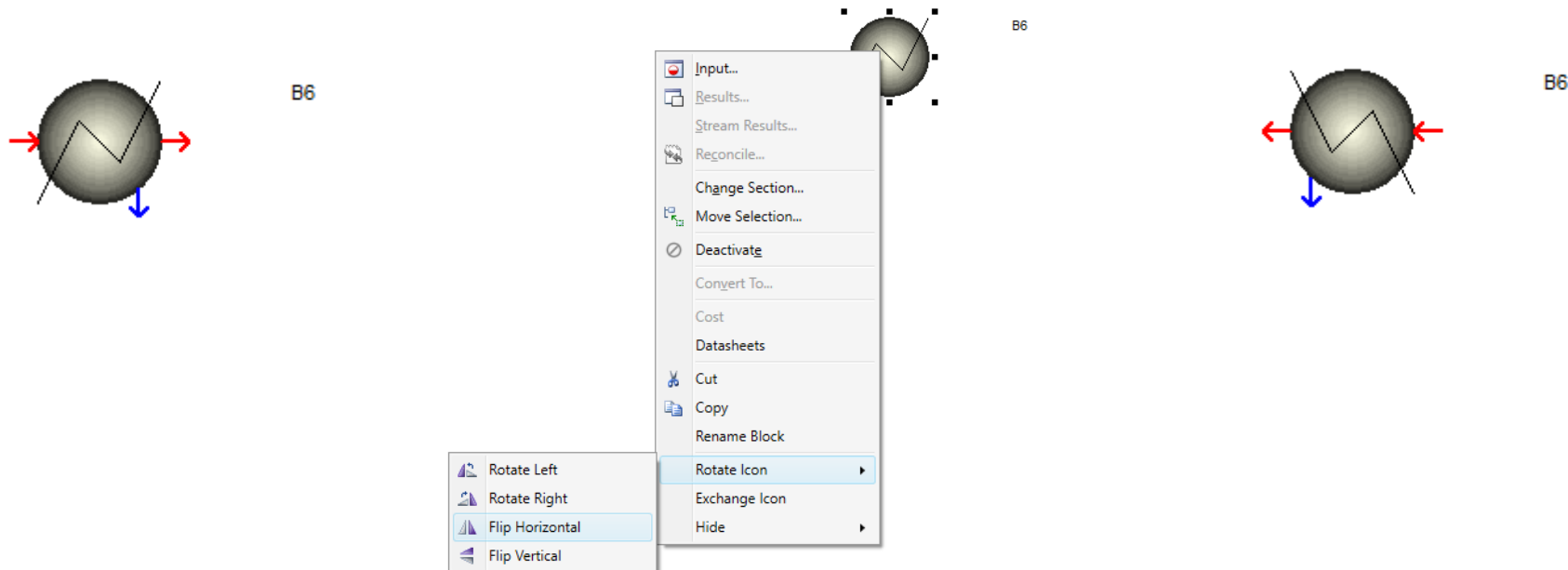
Inserting blocks midstream

- You may also insert a block onto a stream which is already connected to other blocks. This will automatically split and rename the stream into two separate streams



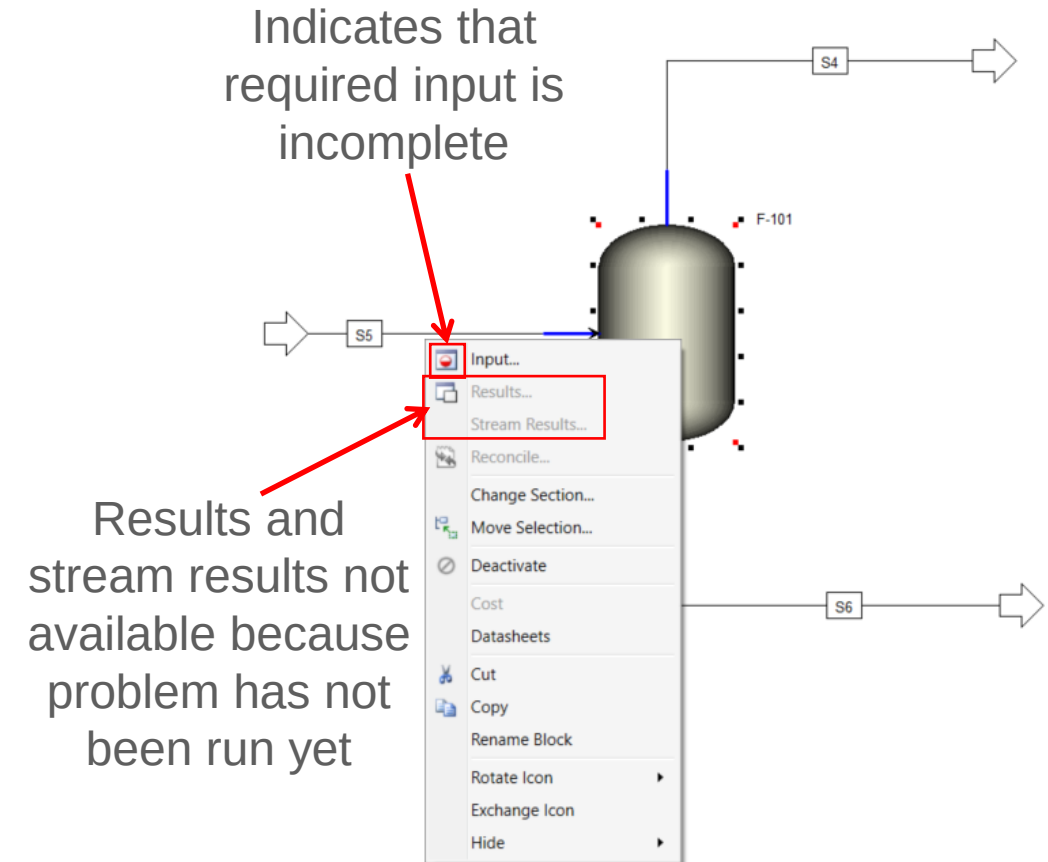
Choosing and rotating icons

- Multiple icons can be used to represent the same block in your flowsheet
- These icons can also be rotated or flipped to change the location and direction of connection ports



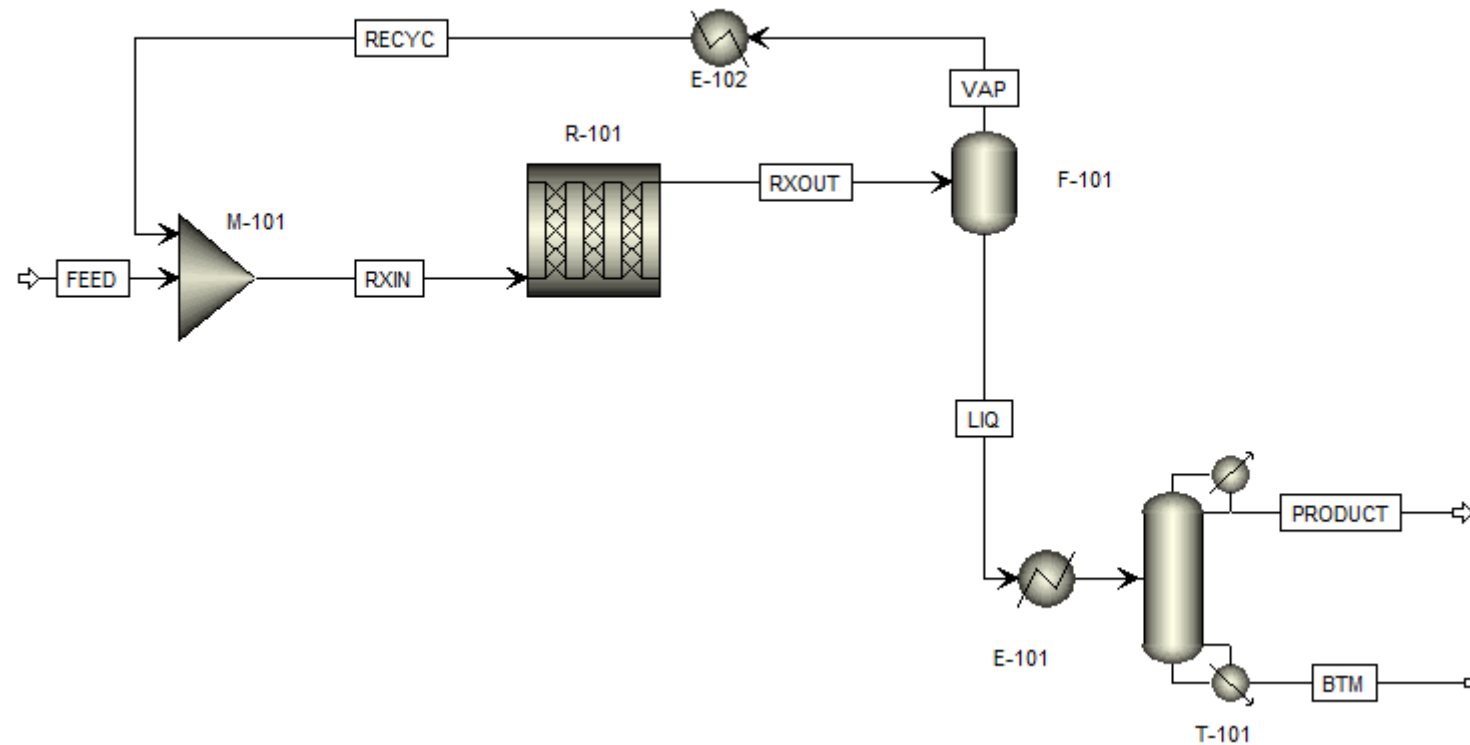
Basic block options

- Basic block options can be accessed by right clicking on the block
- Once you have input the minimum required data and run the problem, the results and stream results can be accessed from here
- You may also rotate the icon, exchange it for another icon from the model palette or hide various tags from the options available



Exercise 2.1

Construct a flowsheet identical to the one shown below.



Separators

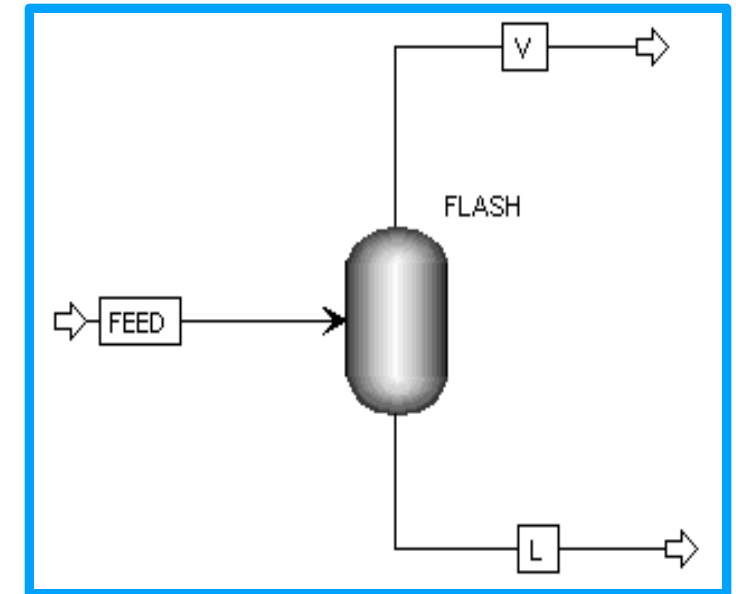
Separator Blocks in Aspen Plus

Flash2	Separates feed into two outlet streams, using rigorous vapor-liquid or vapor-liquid-liquid equilibrium	Flash drums, evaporators, knock-out drums, single stage separators
Flash3	Separates feed into three outlet streams, using rigorous vapor-liquid-liquid equilibrium	Decanters, single-stage separators with two liquid phases
Decanter	Separates feed into two liquid outlet streams	Decanters, single-stage separators with two liquid phases and no vapor phase
Sep	Separates inlet stream components into multiple outlet streams, based on specified flows or split fractions	Component separation operations, such as distillation and absorption, when the details of the separation are unknown or Unimportant
Sep2	Separates inlet stream components into two outlet streams, based on specified flows, split fractions, or purities	Component separation operations, such as distillation and absorption, when the details of the separation are unknown or Unimportant

Flash specifications in Aspen Plus

- Specify fixed operating conditions from the following:
 - Temperature
 - Pressure
 - Vapour Fraction
 - Duty
- The remaining “free” parameters will be calculated

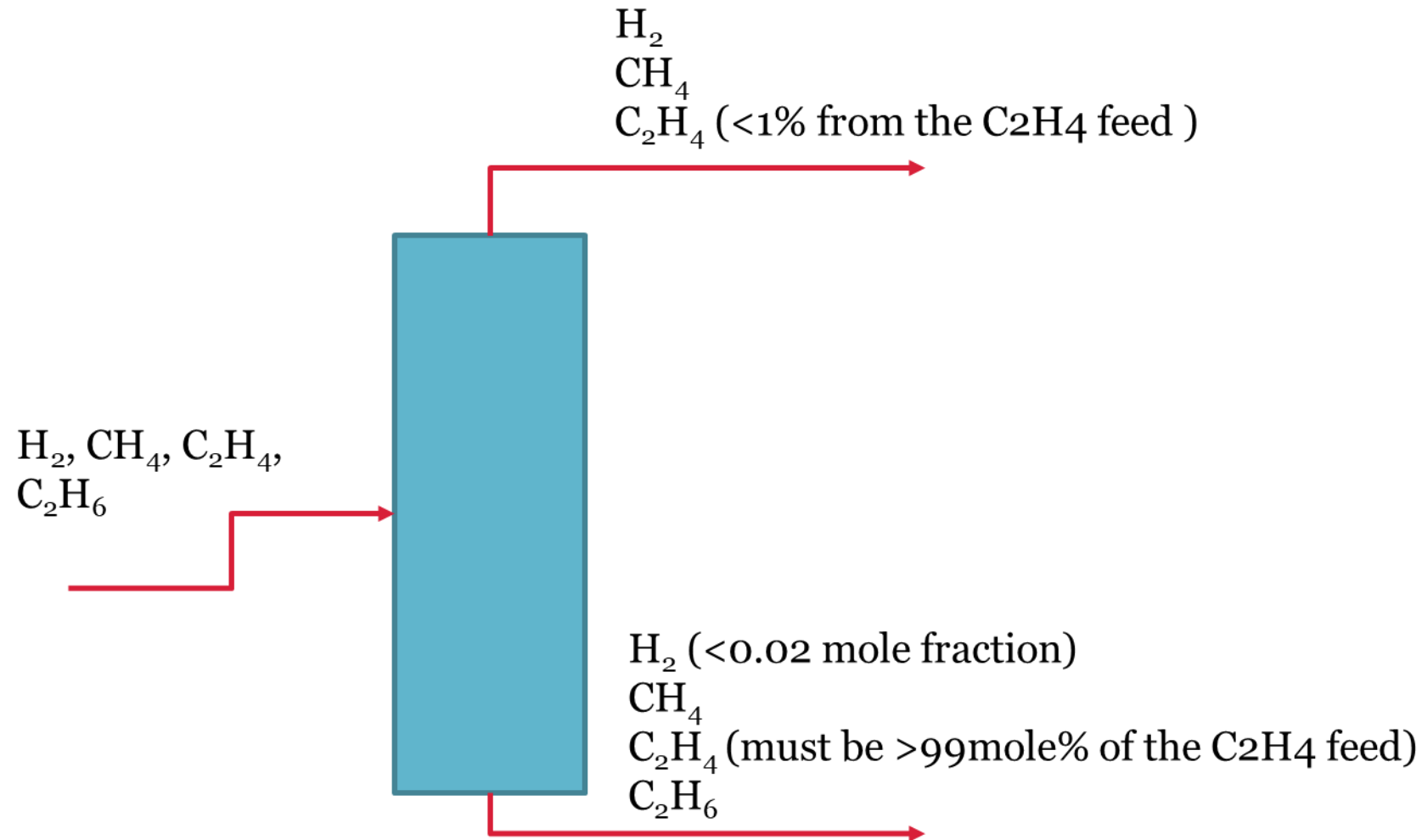
#Specify a suitable thermodynamic property method for accurate vapor-liquid equilibrium data



Example 2.1

- A feed stream in an ethylene plant containing 51.4 mole% hydrogen, 30.2 mole% methane, 15.3 mole% ethylene and 3.1 mole% ethane has a flowrate of 2758.53 kmol/hr at -68°C and 33 bar. Due to the large mole fraction of hydrogen it occupies a large volume and needs to be removed first to reduce the size of subsequent units. To achieve this a flash column operated at 32 bar is used with a vapour fraction equal to the mole fraction of hydrogen in the feed as an initial guess to estimate the heat duty required for this column.
- The flash column needs to operate such that the mole fraction of hydrogen in the bottom stream is less than 2 mole% and the loss of ethylene to the hydrogen stream is less than 1 mole%. RK-Soave can be used for hydrocarbon systems at high pressure.

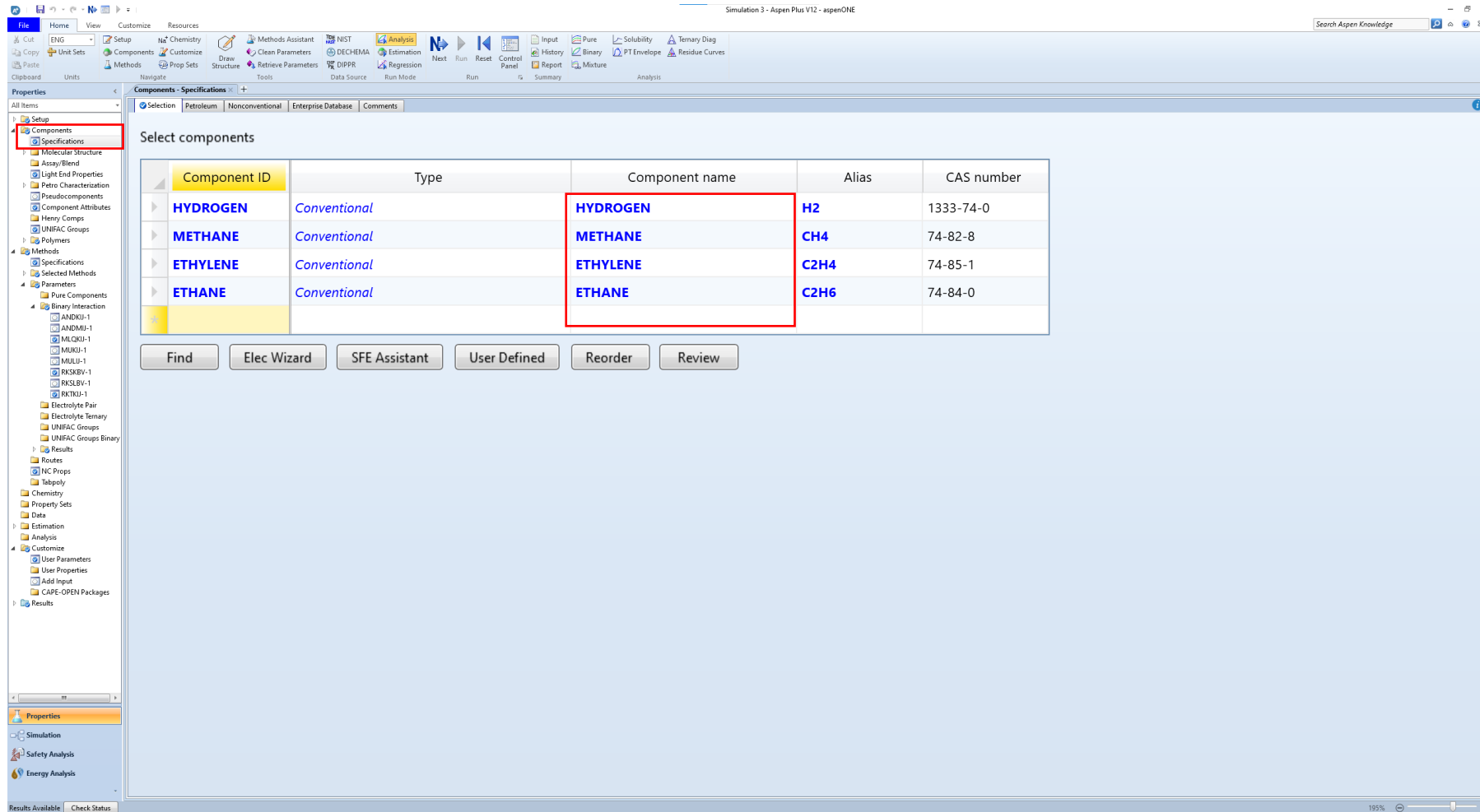
Problem Analysis



Simulation Steps

- Simulation steps:
 - Add the Chemical Components
 - Select the Property Method (Thermodynamics Model)
 - Adding a Flash Drum Unit
 - Specify the User Input
 - Run the Based Case Simulation and Check the Results
 - Perform the Sensitivity Studies to achieve the targeted product specification

Create blank simulation and in the properties environment > component specification > **list down all components involved in process**



Simulation 3 - Aspen Plus V12 - aspenONE

Search Aspen Knowledge

File Home View Customize Resources

ENG Unit Sets Components Customize Draw Structure Methods Assistant Clean Parameters DEHEMA DIPPR Regression Estimation Next Run Reset Control Panel History Binary PT Envelope Mixture Analysis

Properties

Selection Petroleum Nonconventional Enterprise Database Comments

Select components

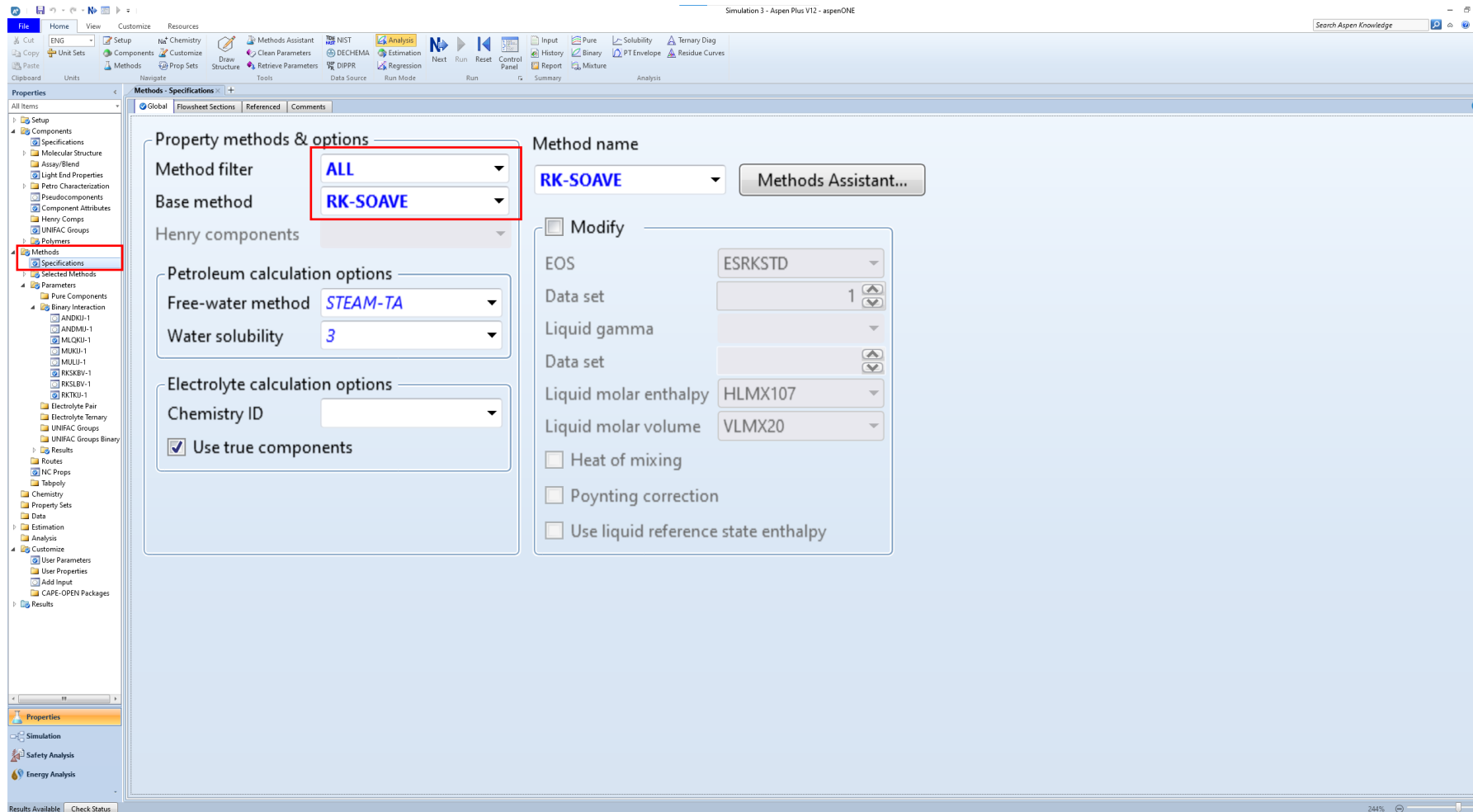
Component ID	Type	Component name	Alias	CAS number
HYDROGEN	Conventional	HYDROGEN	H2	1333-74-0
METHANE	Conventional	METHANE	CH4	74-82-8
ETHYLENE	Conventional	ETHYLENE	C2H4	74-85-1
ETHANE	Conventional	ETHANE	C2H6	74-84-0

Find Elec Wizard SFE Assistant User Defined Reorder Review

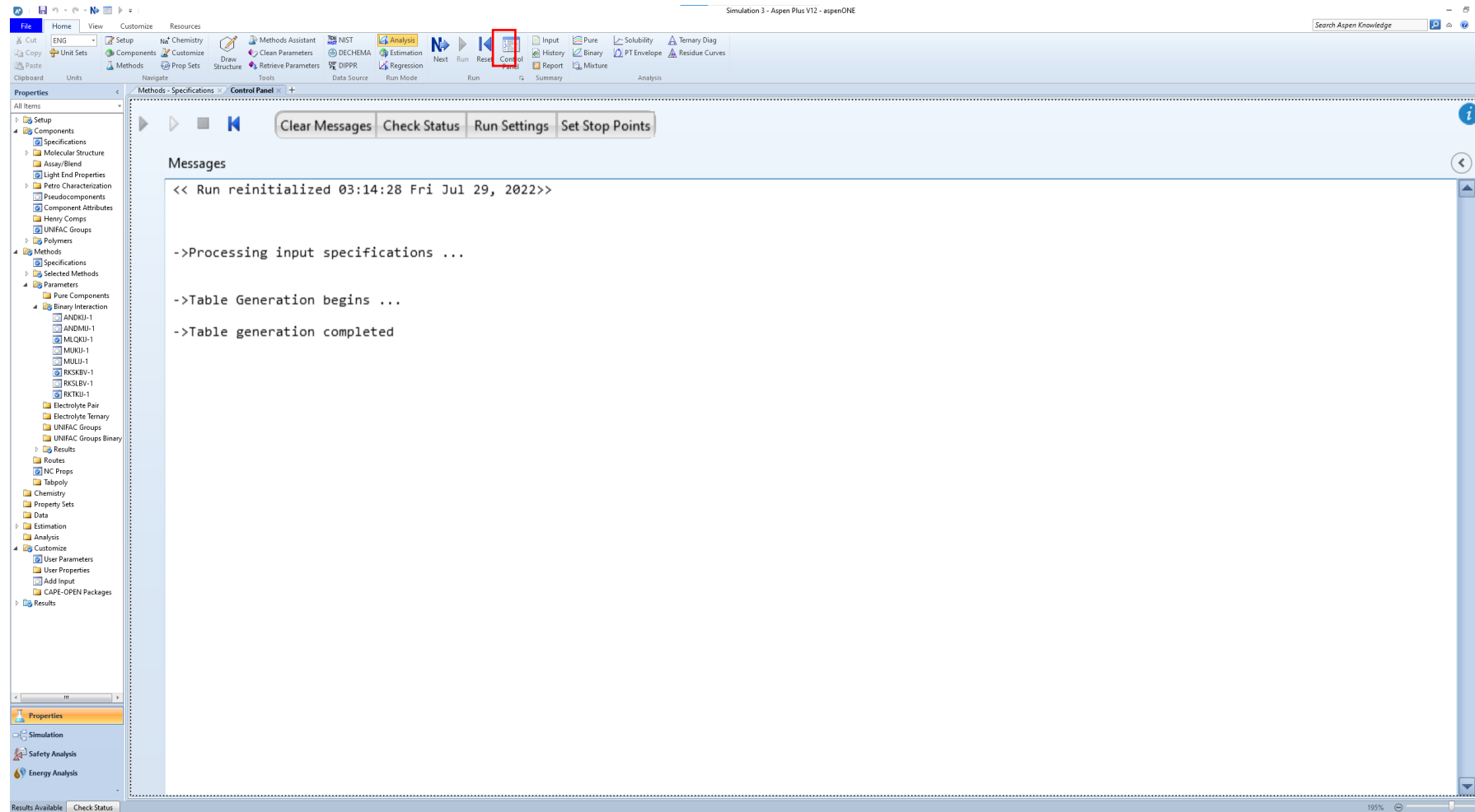
Properties Simulation Safety Analysis Energy Analysis

Results Available Check Status

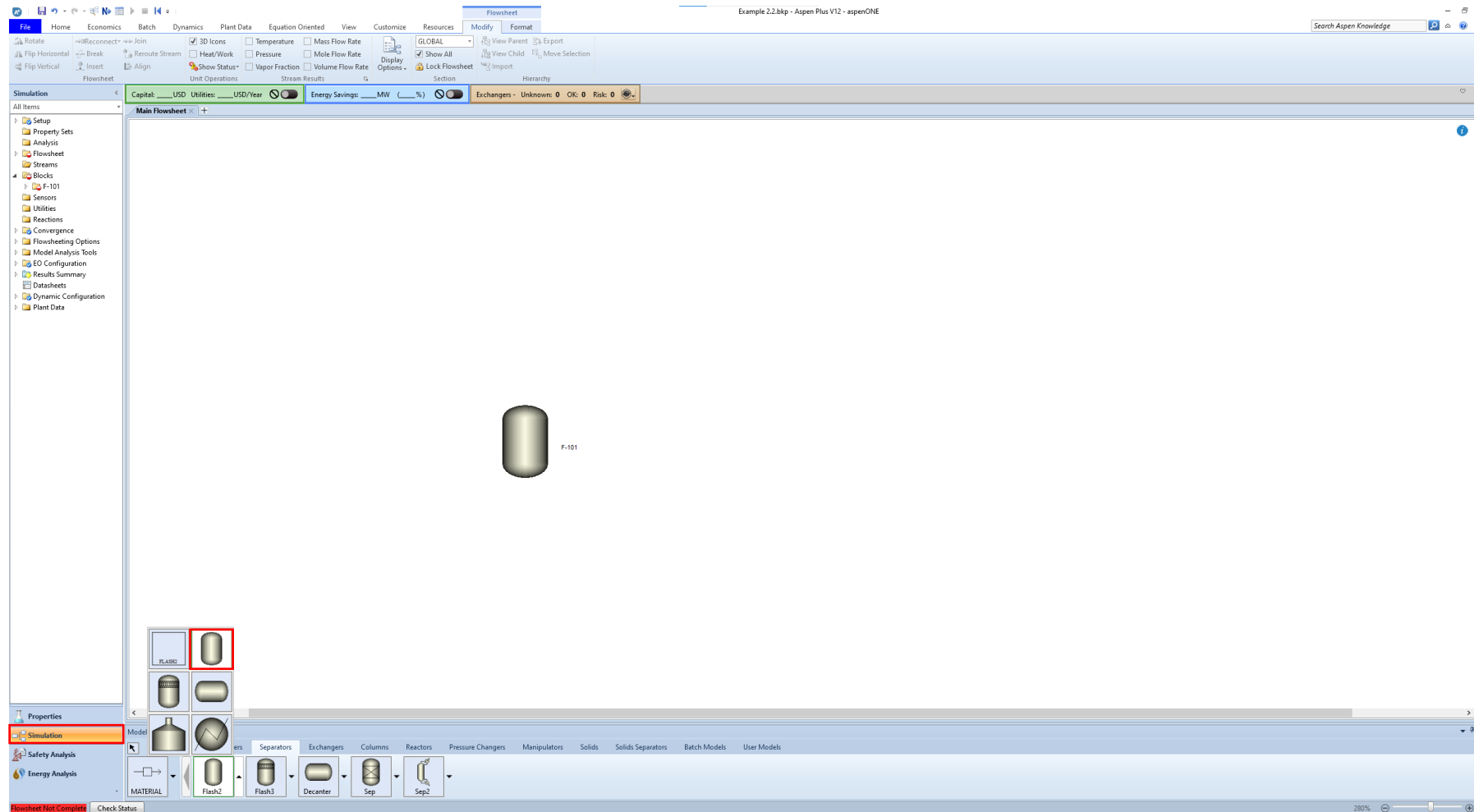
Press next or F4 to move to next required input which will take you to methods specification > **change method filter to all and select RK-Soave**



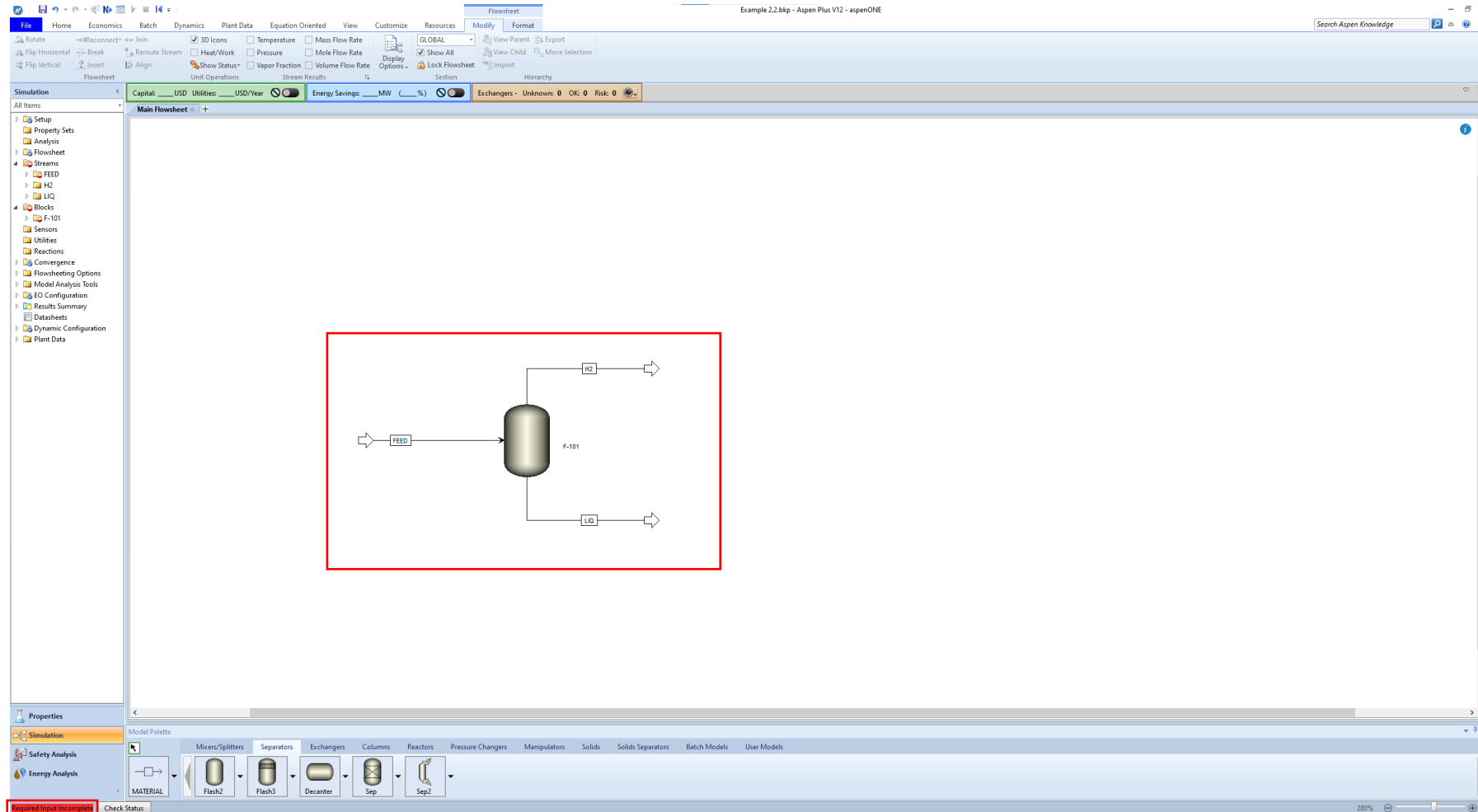
Press run to generate the properties data based on the thermodynamic package selected > **it is good practise to save the simulation file at this juncture**



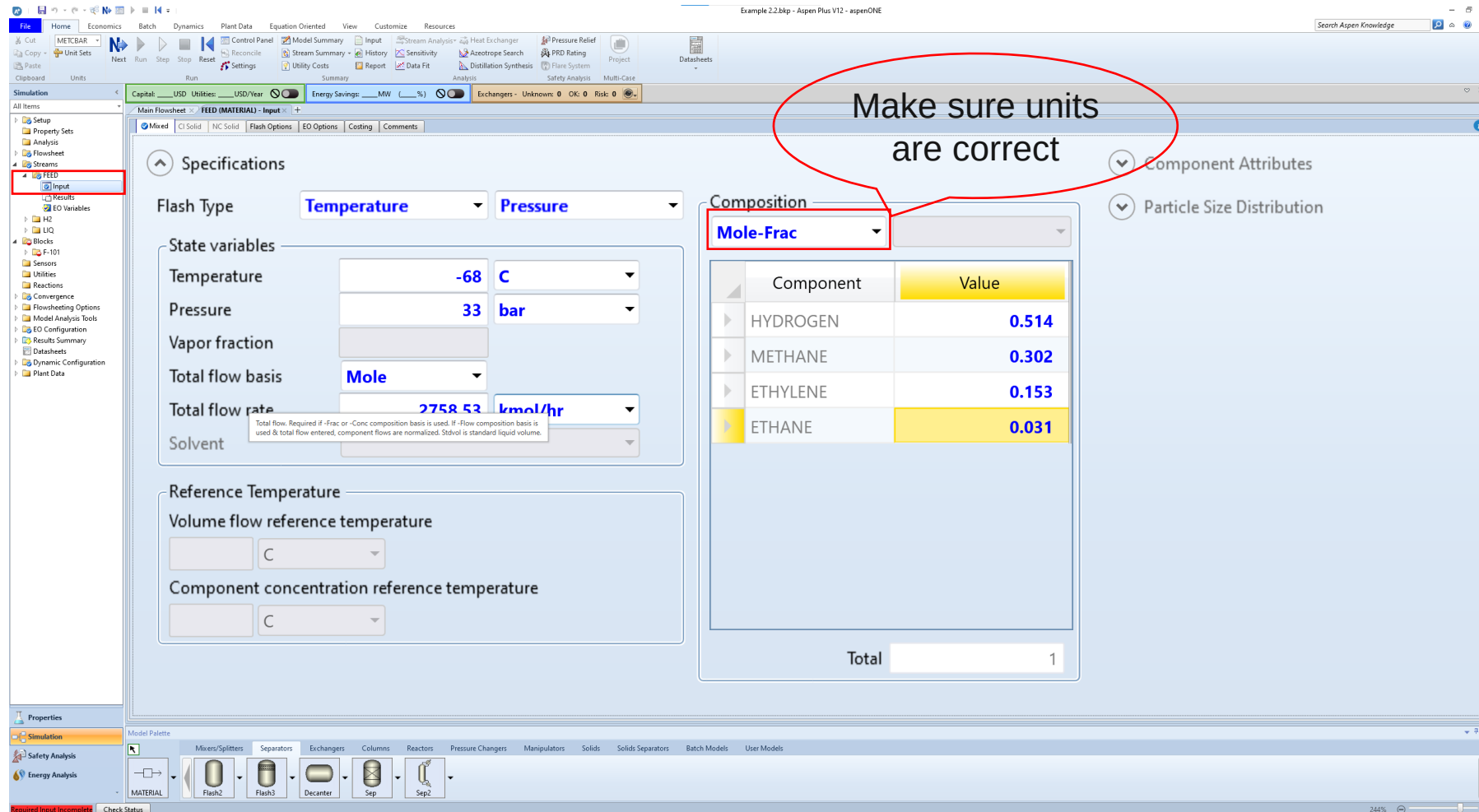
Go to the simulation environment and drag and drop a flash 2 unit onto the main flowsheet



Construct flowsheet by connecting compulsory material streams. Error message will change from flowsheet incomplete to required input incomplete indicating that the flowsheet connections are valid



Press next or F4 to go to feed material input. **Enter feed stream data as provided in the problem**



Example 2.2.bkp - Aspen Plus V12 - aspenONE

Search Aspen Knowledge

Simulation: Capital USD Utilities USD/Year Energy Savings MW % Exchangers Unknowns 0 OK: 0 Risk 0

Main Flowsheet: FEED (MATERIAL) - Input

Specifications

Flash Type: Temperature Pressure

State variables

Temperature: -68 C

Pressure: 33 bar

Vapor fraction

Total flow basis: Mole

Total flow rate: 2758.53 kmol/hr

Solvent

Reference Temperature

Volume flow reference temperature: C

Component concentration reference temperature: C

Composition: Mole-Frac

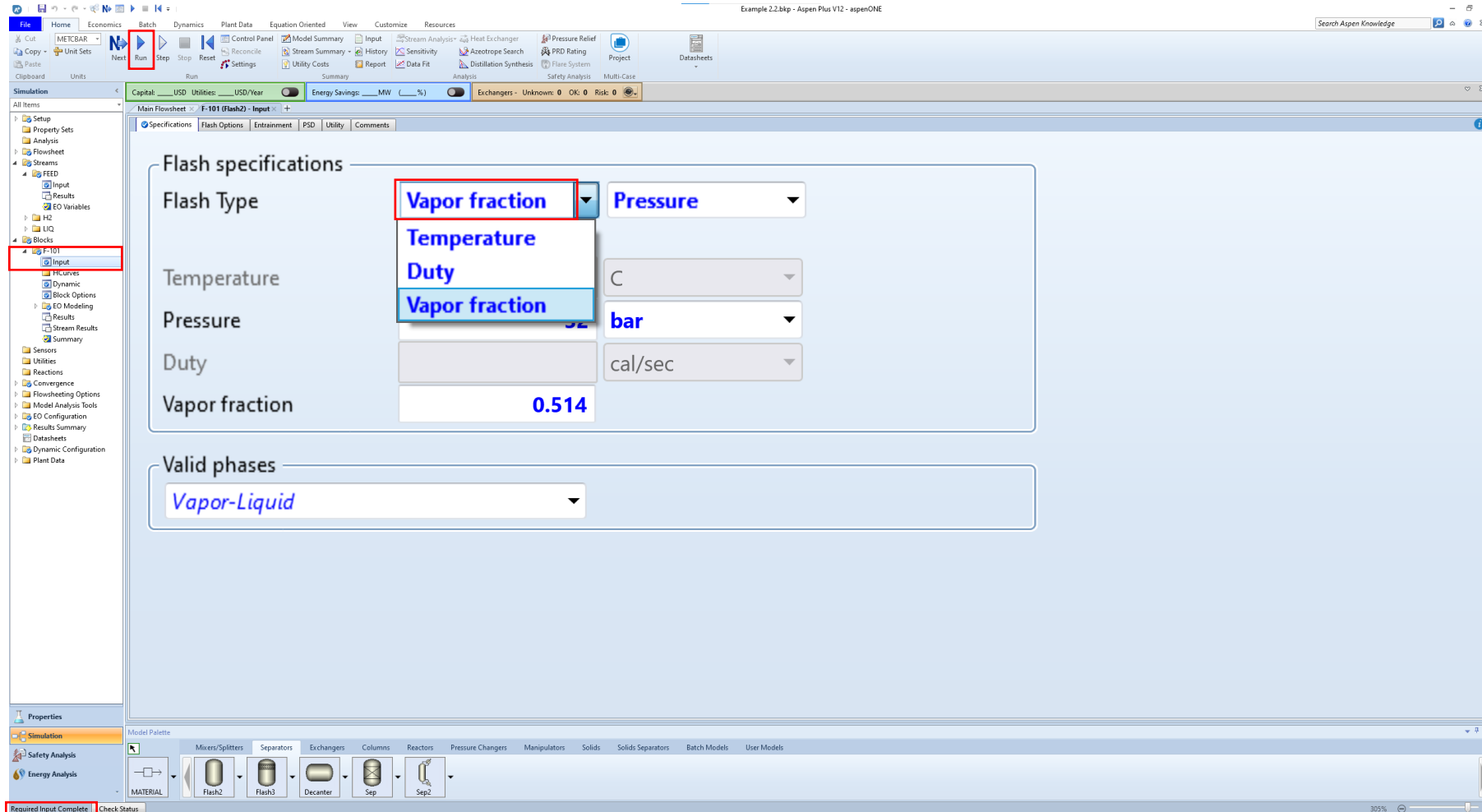
Component	Value
HYDROGEN	0.514
METHANE	0.302
ETHYLENE	0.153
ETHANE	0.031

Total: 1

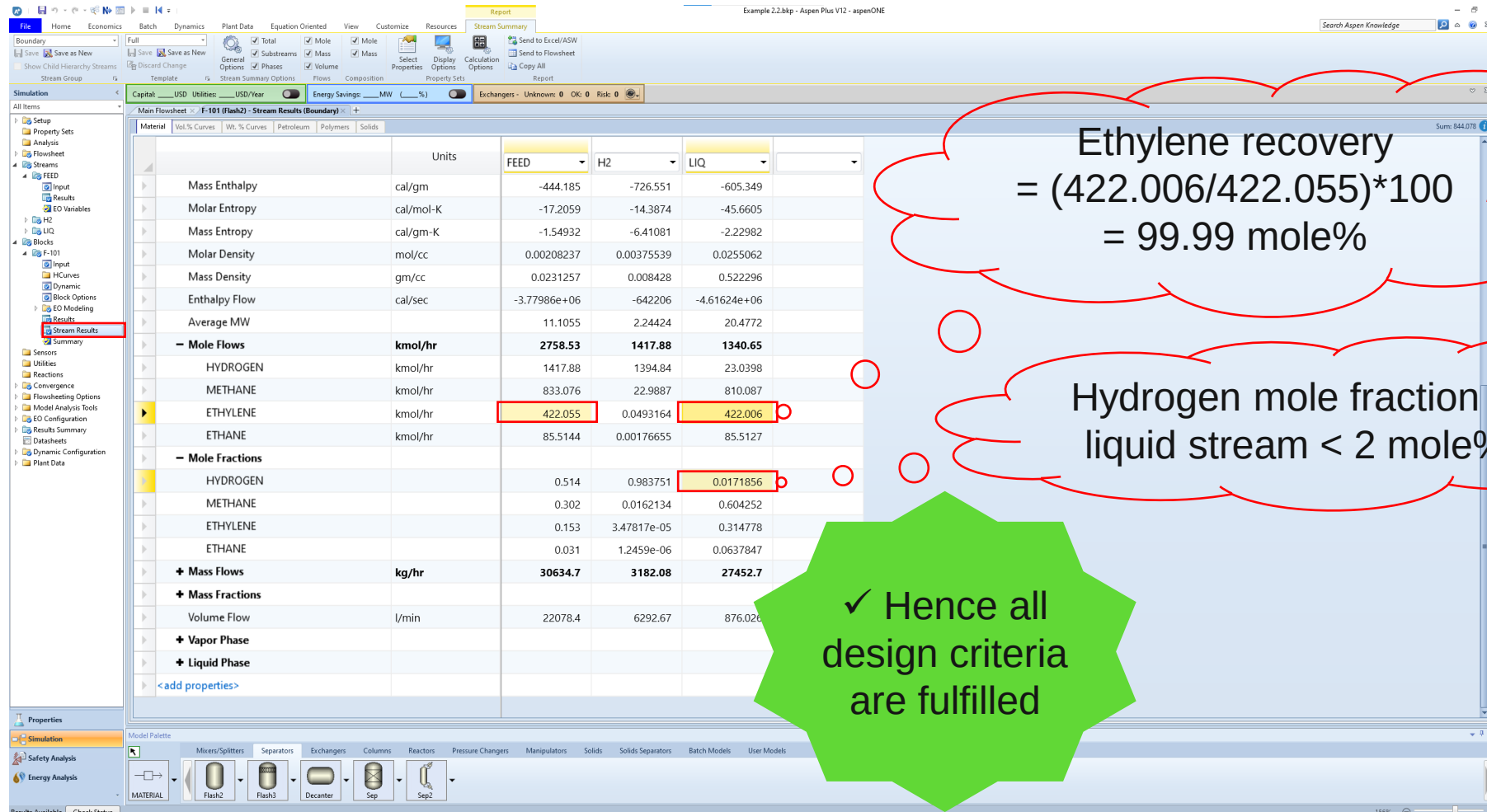
Model Palette: Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids Solids Separators Batch Models User Models

Required Input Incomplete Check Status

Press next or F4 to move to next required input which will bring you to the block input specification > **change the default flash type from temperature to vapour fraction and enter the values from the problem > required input is now complete and simulation can be run**



Go to Blocks > Block name (F-101) > Stream Results or from Main Flowsheet, right click on unit and select Stream Results > **examine results to determine if design objectives were met**



Ethylene recovery

$$= (422.006/422.055) \times 100$$

$$= 99.99 \text{ mole\%}$$

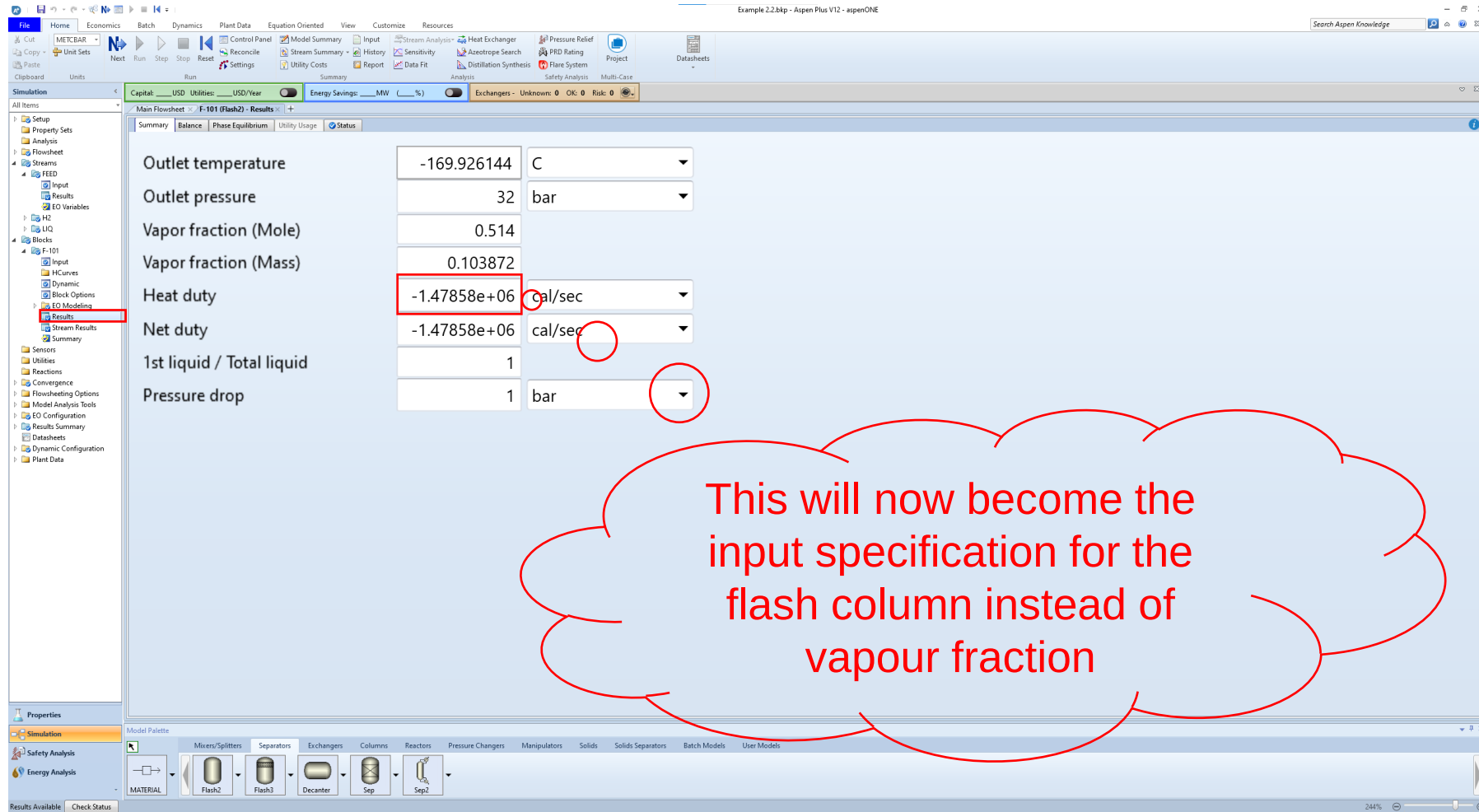
Hydrogen mole fraction in liquid stream < 2 mole%

✓ Hence all design criteria are fulfilled

Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids	Units	FEED	H2	LIQ
Mass Enthalpy						cal/gm	-444.185	-726.551	-605.349
Molar Entropy						cal/mol-K	-17.2059	-14.3874	-45.6605
Mass Entropy						cal/gm-K	-1.54932	-6.41081	-2.22982
Molar Density						mol/cc	0.00208237	0.00375539	0.0255062
Mass Density						gm/cc	0.0231257	0.008428	0.522296
Enthalpy Flow						cal/sec	-3.77986e+06	-642206	-4.61624e+06
Average MW							11.1055	2.24424	20.4772
- Mole Flows						kmol/hr	2758.53	1417.88	1340.65
HYDROGEN						kmol/hr	1417.88	1394.84	23.0398
METHANE						kmol/hr	833.076	22.9887	810.087
ETHYLENE						kmol/hr	422.055	0.0493164	422.006
ETHANE						kmol/hr	85.5144	0.00176655	85.5127
- Mole Fractions									
HYDROGEN							0.514	0.983751	0.0171856
METHANE							0.302	0.0162134	0.604252
ETHYLENE							0.153	3.47817e-05	0.314778
ETHANE							0.031	1.2459e-06	0.0637847
+ Mass Flows						kg/hr	30634.7	3182.08	27452.7
+ Mass Fractions									
Volume Flow						l/min	22078.4	6292.67	876.026
+ Vapor Phase									
+ Liquid Phase									
< add properties >									

Sensitivity Analysis

Now that we have met the design criteria for the flash column we can examine the duty required and attempt to optimize the design of this flash column using sensitivity analysis. **To examine results not provided in the stream table go to Block > Unit name (F-101) > Results or right click the block in the main flowsheet and select results**



Example 2.2.bkp - Aspen Plus V12 - aspenONE

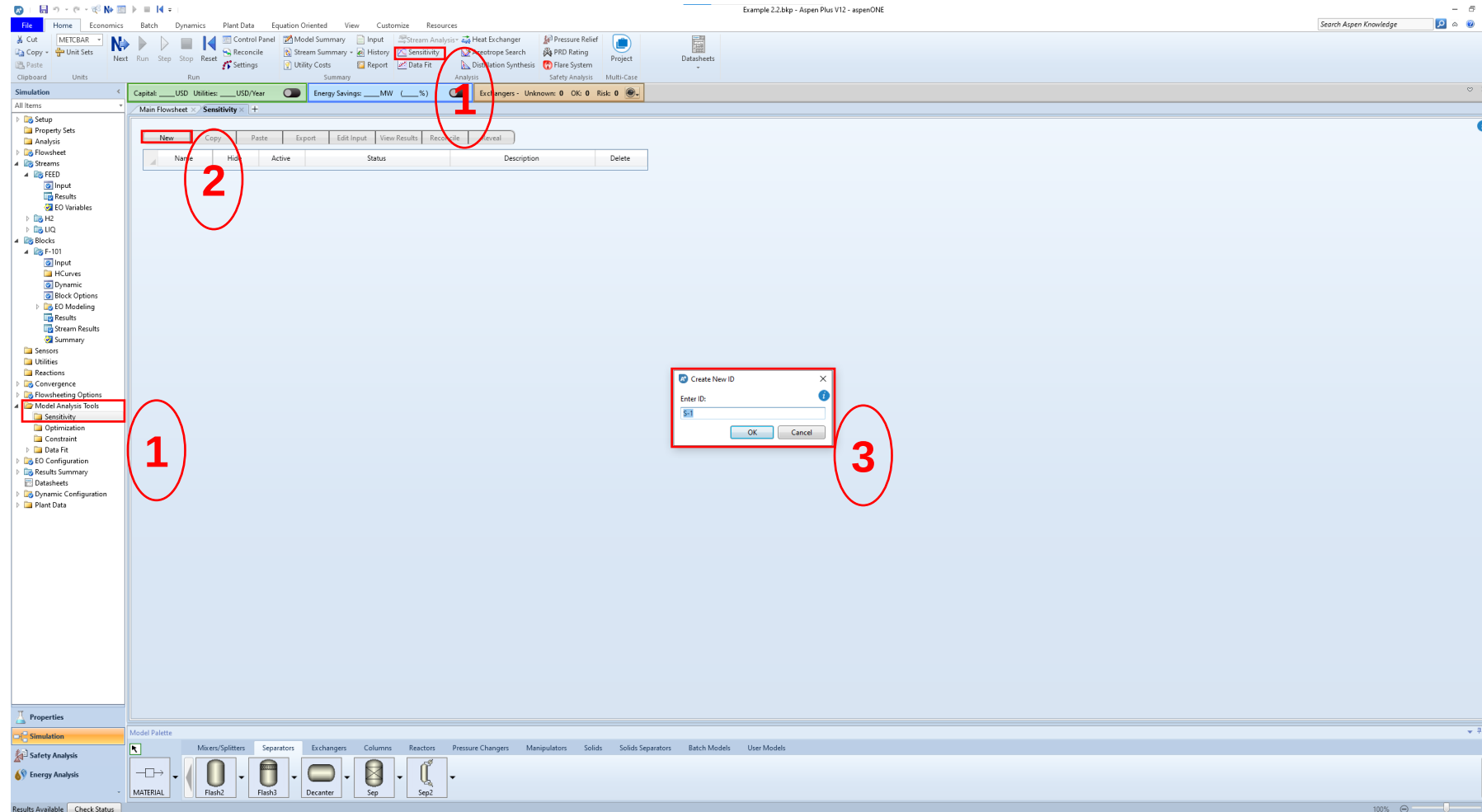
Simulation: Capital: USD Utilities: USD/Year Energy Savings: MW Exchangers: Unknown: 0 OK: 0 Risk: 0

Main Flowsheet: F-101 (Flash2) - Results

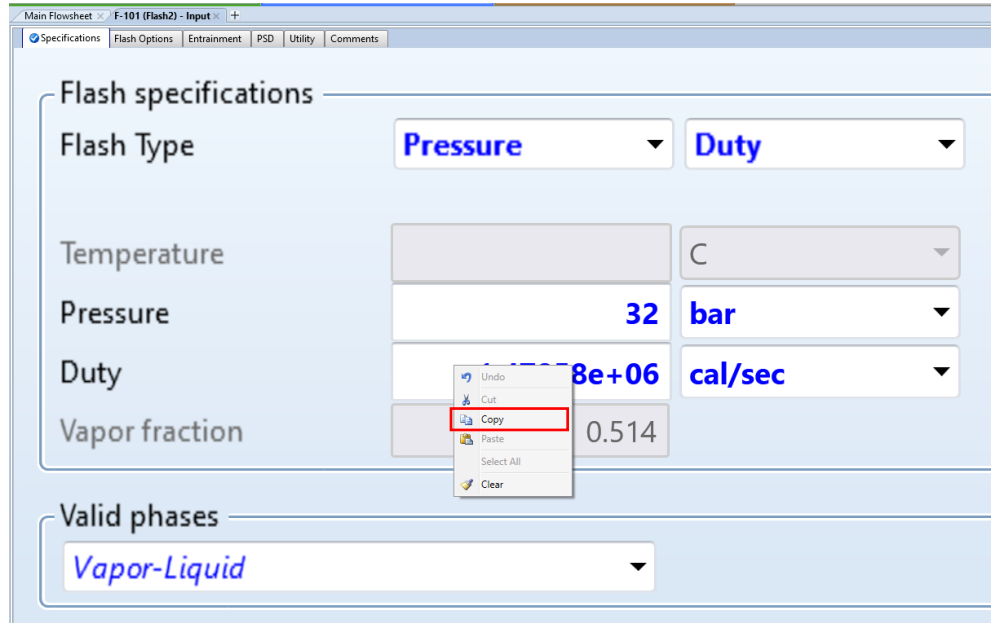
Property	Value	Unit
Outlet temperature	-169.926144	C
Outlet pressure	32	bar
Vapor fraction (Mole)	0.514	
Vapor fraction (Mass)	0.103872	
Heat duty	-1.47858e+06	cal/sec
Net duty	-1.47858e+06	cal/sec
1st liquid / Total liquid	1	
Pressure drop	1	bar

This will now become the input specification for the flash column instead of vapour fraction

1. Either click Sensitivity from the home ribbon or select sensitivity under model analysis tools > 2. Click New > 3. Create a new Sensitivity ID by clicking OK.



Vary refers to the manipulated variable in our analysis (Duty). You can copy and paste it into the form from the flash input specifications tab.



Flash specifications

Flash Type: Pressure, Duty

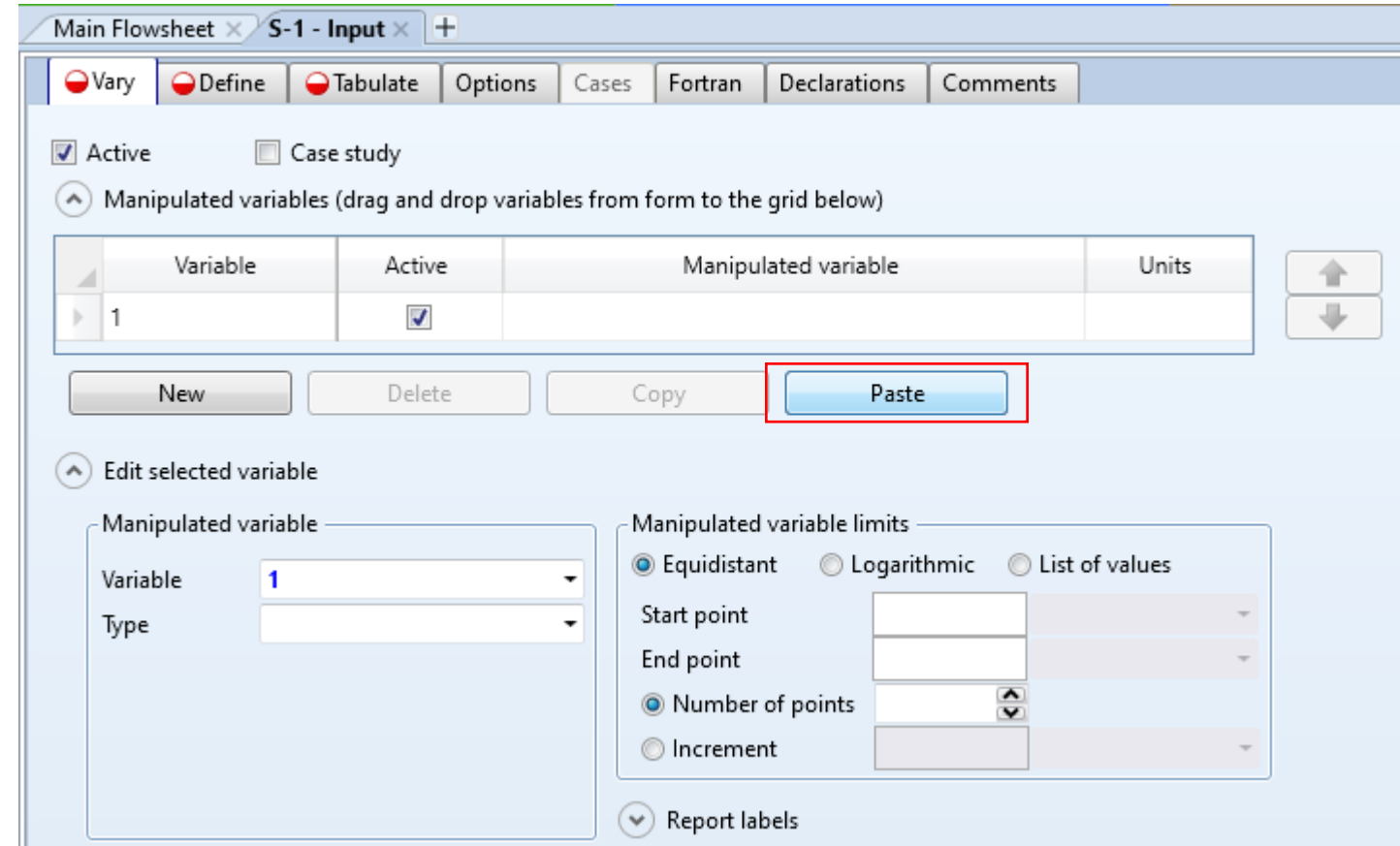
Temperature: C

Pressure: 32 bar

Duty: 8e+06 cal/sec

Vapor fraction: 0.514

Valid phases: Vapor-Liquid



Main Flowsheet x S-1 - Input x +

Vary Define Tabulate Options Cases Fortran Declarations Comments

☒ Active ☐ Case study

Manipulated variables (drag and drop variables from form to the grid below)

Variable	Active	Manipulated variable	Units
1	☒		

New Delete Copy Paste

Edit selected variable

Manipulated variable

Variable: 1

Type:

Manipulated variable limits

☒ Equidistant ☐ Logarithmic ☐ List of values

Start point:

End point:

☒ Number of points:

☐ Increment:

Report labels

Use a suitable range and number of points to begin analysis

Main Flowsheet | S-1 - Results | S-1 - Results Summary - Plot | **S-1 - Input** | S-1 - Results Summary - Plot | +

☒ Vary ☒ Define ☒ Tabulate Options Cases Fortran Declarations Comments

☒ Active ☐ Case study

Manipulated variables (drag and drop variables from form to the grid below)

	Variable	Active	Manipulated variable	Units
1		☒	Block-Var Block=F-101 Variable=DUTY Sentence=P...	cal/sec

New Delete Copy Paste

Edit selected variable

Manipulated variable

Variable: **1**

Type: **Block-Var**

Block: **F-101**

Variable: **DUTY**

Sentence: **PARAM**

Units: **cal/sec**

Manipulated variable limits

☒ Equidistant ☐ Logarithmic ☐ List of values

Start point: **-1.5e6** cal/sec

End point: **-1e6** cal/sec

☒ Number of points: **100**

☐ Increment: cal/sec

Report labels

Similarly to fill the define tab in sensitivity input (dependent variables) the information can be copied from the stream results tab and pasted in the sensitivity input > define tab

Main Flowsheet - F-101 (Flash2) - Stream Results (Boundary)					
Material	Vol. % Curves	Wt. % Curves	Petroleum	Polymers	Solids
	Units	FEED	H2	LIQ	
Description					
From			F-101	F-101	
To		F-101			
Stream Class		CONVEN	CONVEN	CONVEN	
Maximum Relative Error					
Cost Flow	\$/hr				
- MIXED Substream					
Phase			Vapor Phase	Liquid Phase	
Temperature	C	-68	-169.926	-169.926	
Pressure	bar	33	32	32	
Molar Vapor Fraction		0.990347	1	0	
Molar Liquid Fraction		0.00965339	0	1	
Molar Solid Fraction		0	0	0	
Mass Vapor Fraction		0.977264	1	0	
Mass Liquid Fraction		0.0227361	0	1	
Mass Solid Fraction		0	0	0	
Molar Enthalpy	cal/mol	-4932.88	-1630.56	-12395.9	
Mass Enthalpy	cal/gm	-444.185	-726.551	-605.349	
Molar Entropy	cal/mol-K	-17.2059	-14.3874	-45.6605	
Mass Entropy	cal/gm-K	-1.54932	-6.41081	-2.22982	
Molar Density	mol/cc	0.00208237	0.00375539	0.0255062	
Mass Density	gm/cc	0.0231257	0.008428	0.522296	
Enthalpy Flow	cal/sec	-3.77986e+06	-642206	-4.61624e+06	
Average MW		11.1055	2.24424	20.4772	
- Mole Flows					
	kmol/hr	2758.53	1417.88	1340.65	
HYDROGEN	kmol/hr	1417.88	1394.84	23.0398	
METHANE	kmol/hr	833.076	22.9887	810.087	
ETHYLENE	kmol/hr	422.055	0.0493164	422.006	
ETHANE	kmol/hr	85.5144	0.00176655	85.5127	
- Mole Fractions					
HYDROGEN		0.514	0.990347	0.0174956	
METHANE		0.302			
ETHYLENE		0.153			
ETHANE		0.031	1.2459e-06	0.0637847	
+ Mass Flows					
	kg/hr	30634.7	3182.08	27452.7	
+ Mass Fractions					
Volume Flow	l/min	22078.4	6292.67	876.026	
+ Vapor Phase					

Main Flowsheet S-1 - Input

Vary Define Tabulate Options Cases Fortran Declarations Comments

Sampled variables (drag and drop variables from form to the grid below)

Variable	Definition
MOLEFRA	Mole-Frac Stream=LIQ Substream=MIXED Component=HYDROGEN

New Delete Copy Paste Move Up Move Down View Variables

Edit selected variable

Variable MOLEFRA

Category

- All
- Blocks
- Streams
- Model Utility
- Property Parameters
- Reactions

Reference

Type Mole-Frac

Stream: LIQ

Substream: MIXED

Component: HYDROGEN

This is repeated for the ethylene flowrate in the feed and ethylene flowrate in the bottom product of the flash column. On the next page click fill variables and enter the formula for ethylene recovery in the liquid stream.

Main Flowsheet x S-1 - Results x S-1 - Results Summary - Plot x **S-1 - Input** x S-1 - Results Summary - Plot x

☒ Vary ☒ Define ☒ Tabulate Options Cases Fortran Declarations Comments

Sampled variables (drag and drop variables from form to the grid below)

Variable	Definition
H2MOLFR	Mole-Frac Stream=LIQ Substream=MIXED Component=HYDROGEN
ETYFEDFL	Mole-Flow Stream=FEED Substream=MIXED Component=ETHYLENE Units=kmol/hr
ETYBTMFL	Mole-Flow Stream=LIQ Substream=MIXED Component=ETHYLENE Units=kmol/hr

New Delete Copy Paste Move Up Move Down View Variables

Edit selected variable

Variable **H2MOLFR**

Category

- ☒ All
- ☐ Blocks
- ☐ Streams
- ☐ Model Utility
- ☐ Property Parameters
- ☐ Reactions

Reference

Type: **Mole-Frac**

Stream: **LIQ**

Substream: **MIXED**

Component: **HYDROGEN**

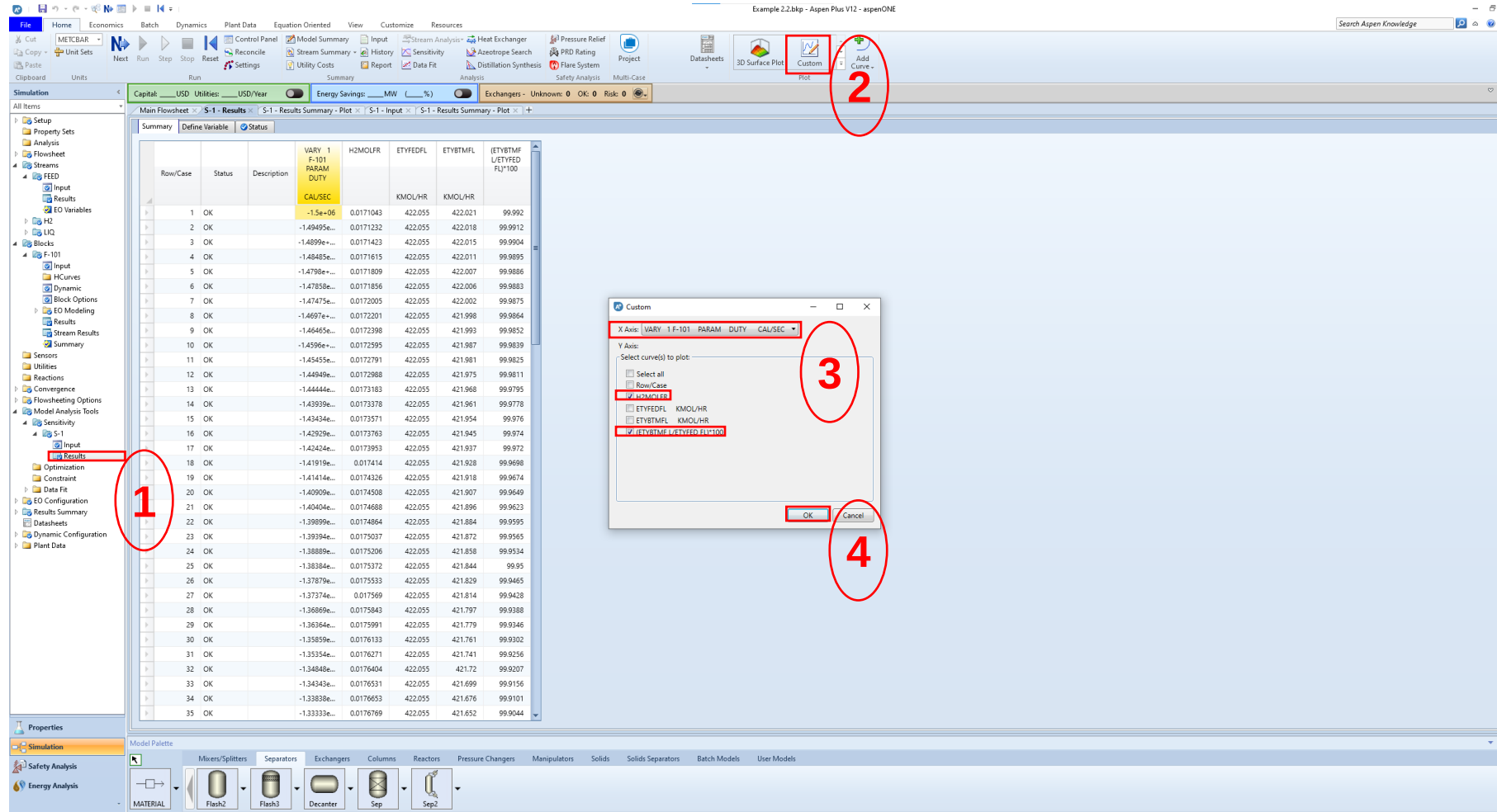
Main Flowsheet x S-1 - Results x S-1 - Results Summary - Plot x **S-1 - Input** x S-1 - Results Summary - Plot x

☒ Vary ☒ Define ☒ Tabulate Options Cases Fortran Declarations Comments

Fill Variables Table Format

Column No	Tabulated variable or expression
1	H2MOLFR
2	ETYFEDFL
3	ETYBTMFL
4	$(\text{ETYBTMFL} / \text{ETYFEDFL}) * 100$

1) Run the Sensitivity analysis and go to the results > 2. select custom plot > 3. Select Vary as X Axis > 4. Select H2MOLFR and Ethylene recovery and press OK to generate plot



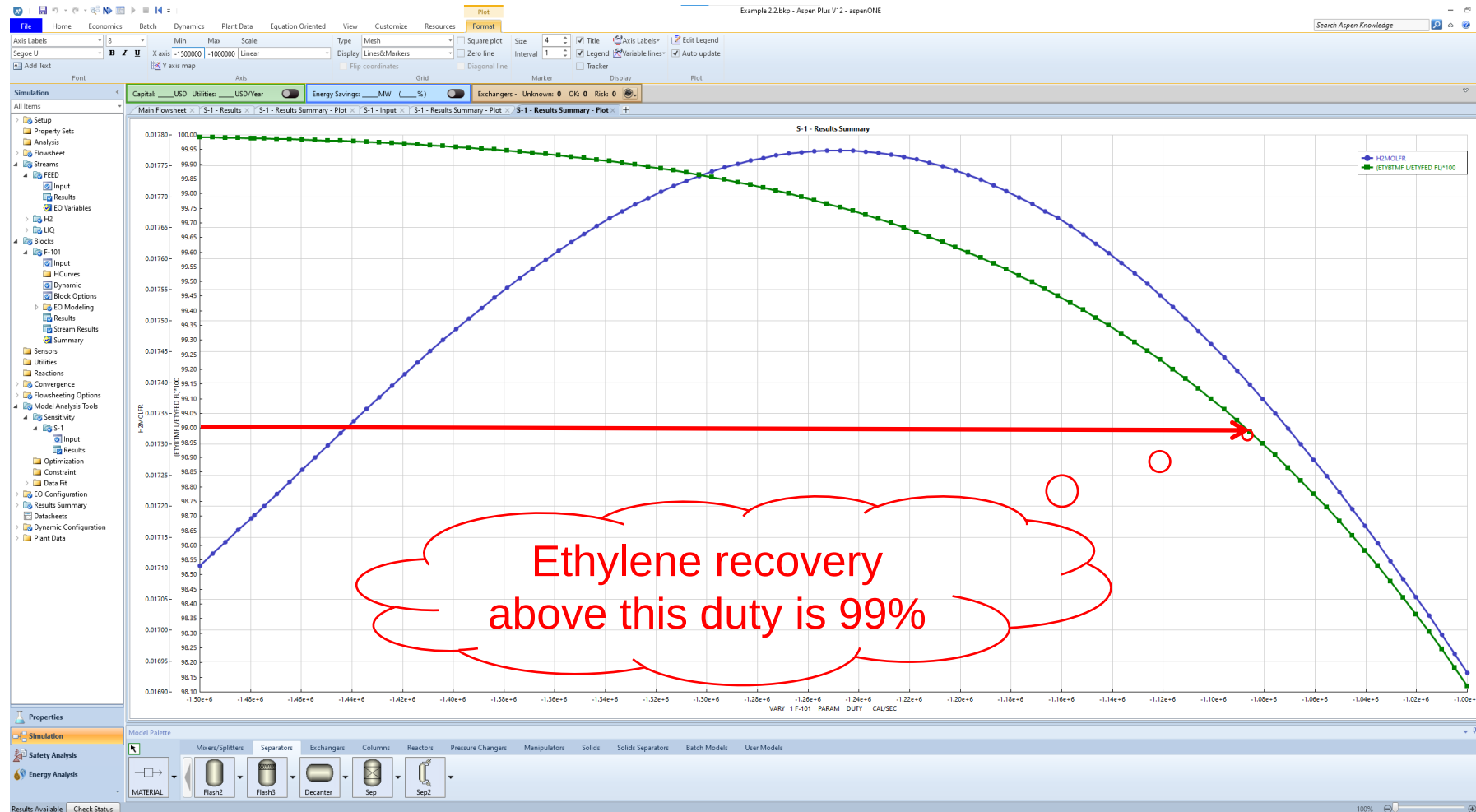
The screenshot displays the Aspen Plus V12 software interface. The main window shows the 'S-1 - Results' tab, which contains a table of simulation results. The table has columns for 'Row/Case', 'Status', 'Description', 'VARY 1 F-101 PARAM DUTY CAL/SEC', 'H2MOLFR', 'ETYFDFL', 'ETYBTMFL', and 'ETYBTMFL U/ETYFDFL*100'. The 'Description' column lists various process parameters and their values. The 'Status' column indicates 'OK' for all rows. The 'VARY 1 F-101 PARAM DUTY CAL/SEC' column shows values ranging from -1.5e+06 to -1.33333e+06. The 'H2MOLFR' column shows values ranging from 0.0171043 to 0.0176769. The 'ETYFDFL' column shows values ranging from 422.055 to 422.055. The 'ETYBTMFL' column shows values ranging from 422.021 to 421.652. The 'ETYBTMFL U/ETYFDFL*100' column shows values ranging from 99.992 to 99.9044.

On the left side of the interface, the 'Sensitivity' tool is highlighted in the 'Tools' pane. The 'Results' sub-tool is also highlighted. The 'Custom' dialog box is open, showing the 'X Axis' as 'VARY 1 F-101 PARAM DUTY CAL/SEC'. The 'Y Axis' is set to 'H2MOLFR' and 'ETYBTMFL U/ETYFDFL*100'. The 'OK' button is highlighted.

Red circles and numbers are used to highlight the steps in the process:

- 1: The 'Sensitivity' tool in the 'Tools' pane.
- 2: The 'Custom' button in the 'Sensitivity' toolbar.
- 3: The 'X Axis' dropdown menu in the 'Custom' dialog box.
- 4: The 'OK' button in the 'Custom' dialog box.

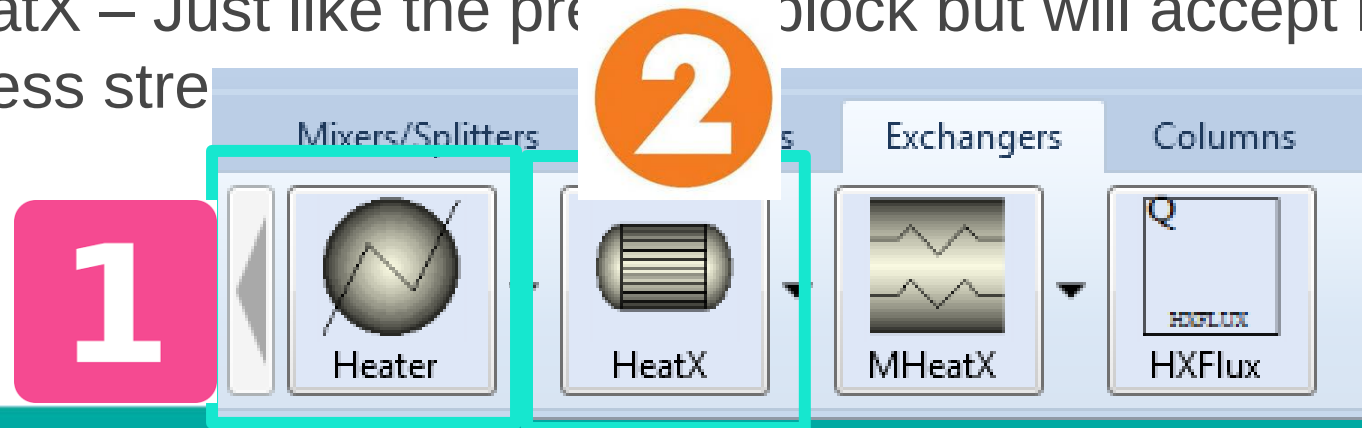
The generated plot shows the heat duty (x-axis) versus the design variables (y-axis). Hydrogen mole fraction in liquid stream is below 2% for entire range tested.



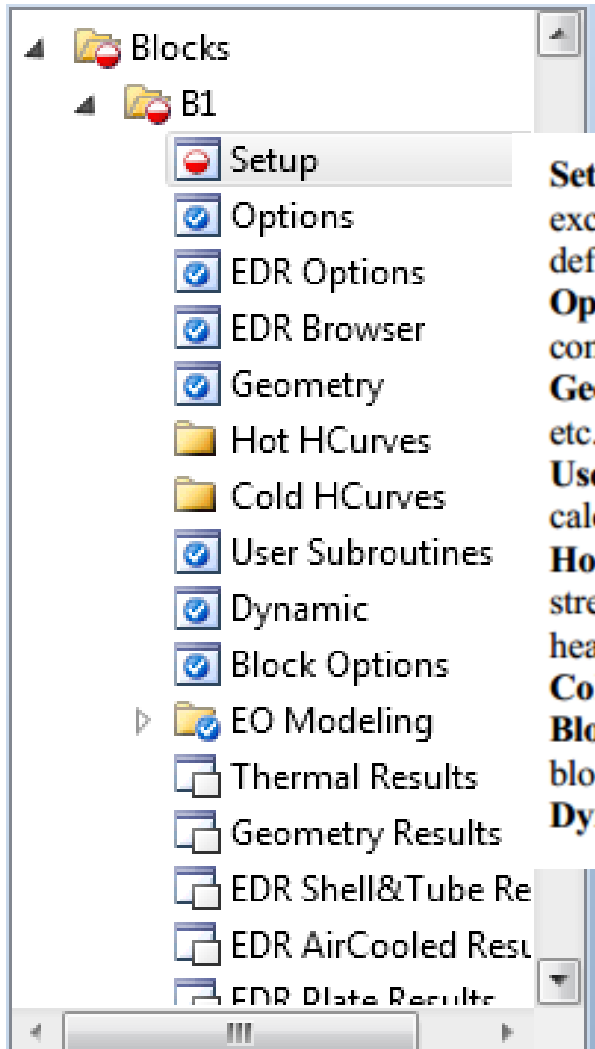
Minimum heat duty

Exchanger Blocks in Aspen Plus

- Heater – Basic heat exchanger, it performs simple energy balance calculations; requires only **ONE** process stream
- HeatX – Fundamental heat exchanger algorithms, used in rigorous design, will calculate energy balances, pressure drops, exchanger area, velocities, etc.; requires **TWO** process streams—a hot and a cold. This block will be used for our design calculations.
- MheatX – Just like the previous block but will accept more process streams



HeatX Setup (Overview)



Setup—(shown in picture) this page shows the calculation type: shortcut or detailed, the exchanger specification, whether the exchanger is countercurrent or cocurrent, and defines how Aspen will calculate the heat transfer coefficients.

Options—shows the valid phases for each stream (e.g. Vapor-liquid), also defines the convergence for calculations

Geometry—shows the input pages for the exchanger layout: tubes, diameters, baffles, etc. *Used only in the detailed calculations.*

User Subroutine—provides an interface so the user can create an algorithm to execute calculations for design (involves FORTRAN programming)

Hot H-curves—shows input pages so that Aspen will create enthalpy profiles for the hot stream, very useful for boiling/condensation processes and provides some insight on the heat transfer

Cold H-curves—same as the Hot H-curves, but creates profiles for the cold stream

Block Options—pages that show the property methods, and simulation options for the block

Dynamic—used only for dynamic mode, not steady-state

Exchanger Specifications (Options)

Hot/Cold stream outlet temperature—specify the outlet temperature of one of the streams, *used for situations where there is not any phase change*

Hot/Cold stream temperature change—specify the temperature increase or decrease of one of the streams

Hot outlet temperature approach—specify the temperature difference between the hot stream outlet temperature and the cold stream inlet temperature, *used with countercurrent flow*

Hot/Cold stream degrees superheat/subcool—specify the outlet temperature below the dew point or above the boiling point of a certain stream, *used in boiling and condensation*

Hot/Cold stream vapor fraction—specify the outlet vapor fraction for a certain stream (1.0 = sat. vapor and 0.0 = sat. liquid), *used for boiling and condensation design*

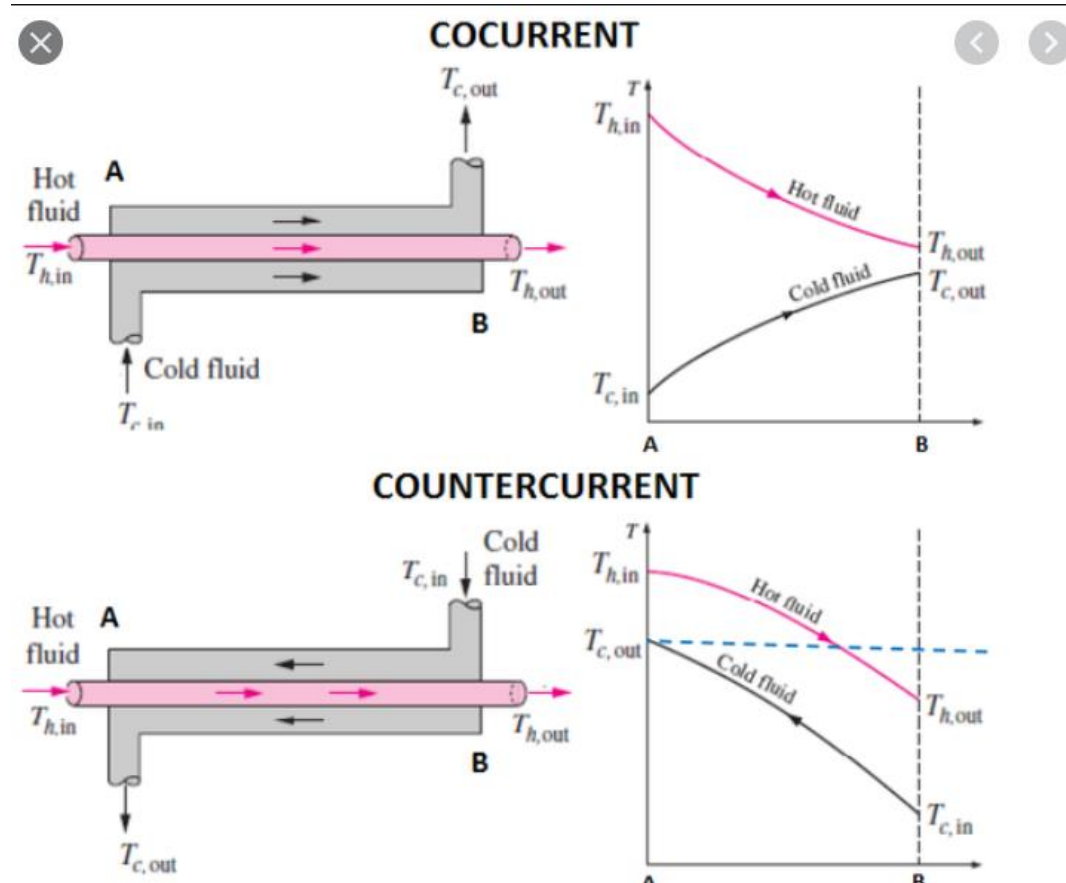
Cold outlet temperature approach—specify the temperature difference between the cold stream outlet temperature and the hot stream inlet temperature, *used with countercurrent flow*

Heat Transfer Area—specify the area of the exchanger, *good for problems where the size of the heat exchanger is fixed*

Heat Duty—specify the amount of energy transferred from one stream to another

Geometry—Calculations based on exchanger layout, *good for use as a check on calculations* (Note*, you need to choose the detailed calculation type to access the geometry option)

Minimum temperature approach



Example 2.2

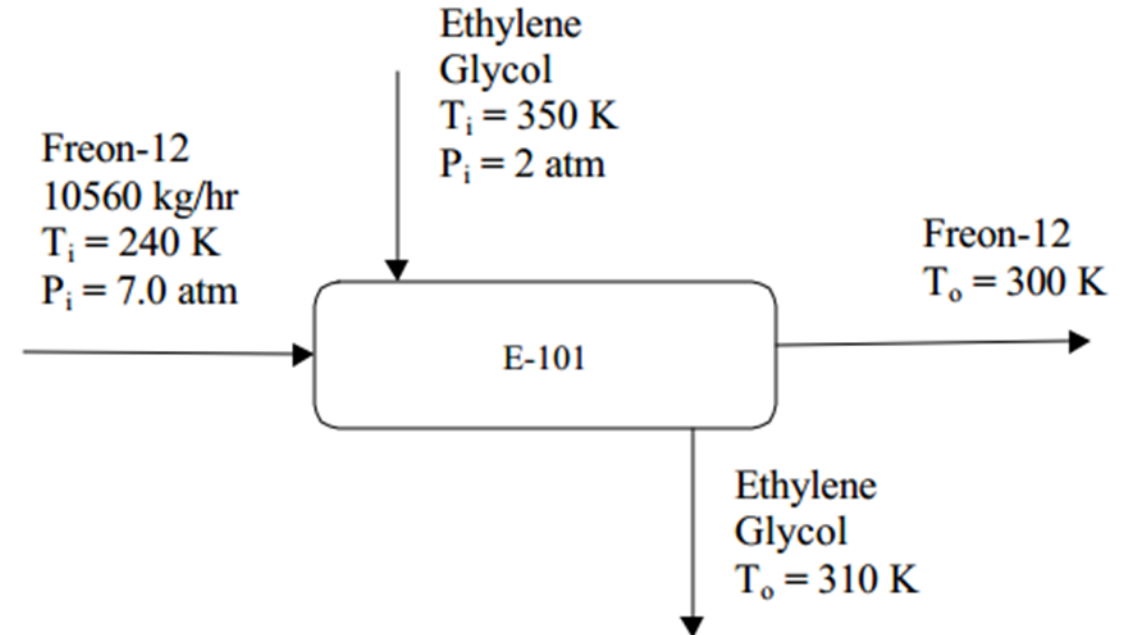
Freon-12 (Dichlorodifluoromethane), at a flow rate of 10560 kg/hr, needs to be heated from 240 K to 300 K. Ethylene glycol is available at 350 K. A typical co-current shell and tube heat exchanger will be used. The plant manager recommends that the minimum temperature approach should be at least 10 K. He also recommends using carbon-steel tubing with a pressure drop not exceeding 10 psig (0.67 atm) on **the shell side** which carries the ethylene glycol.

Simulate the heat exchange process using **Heater model** and determine the heat duty required.

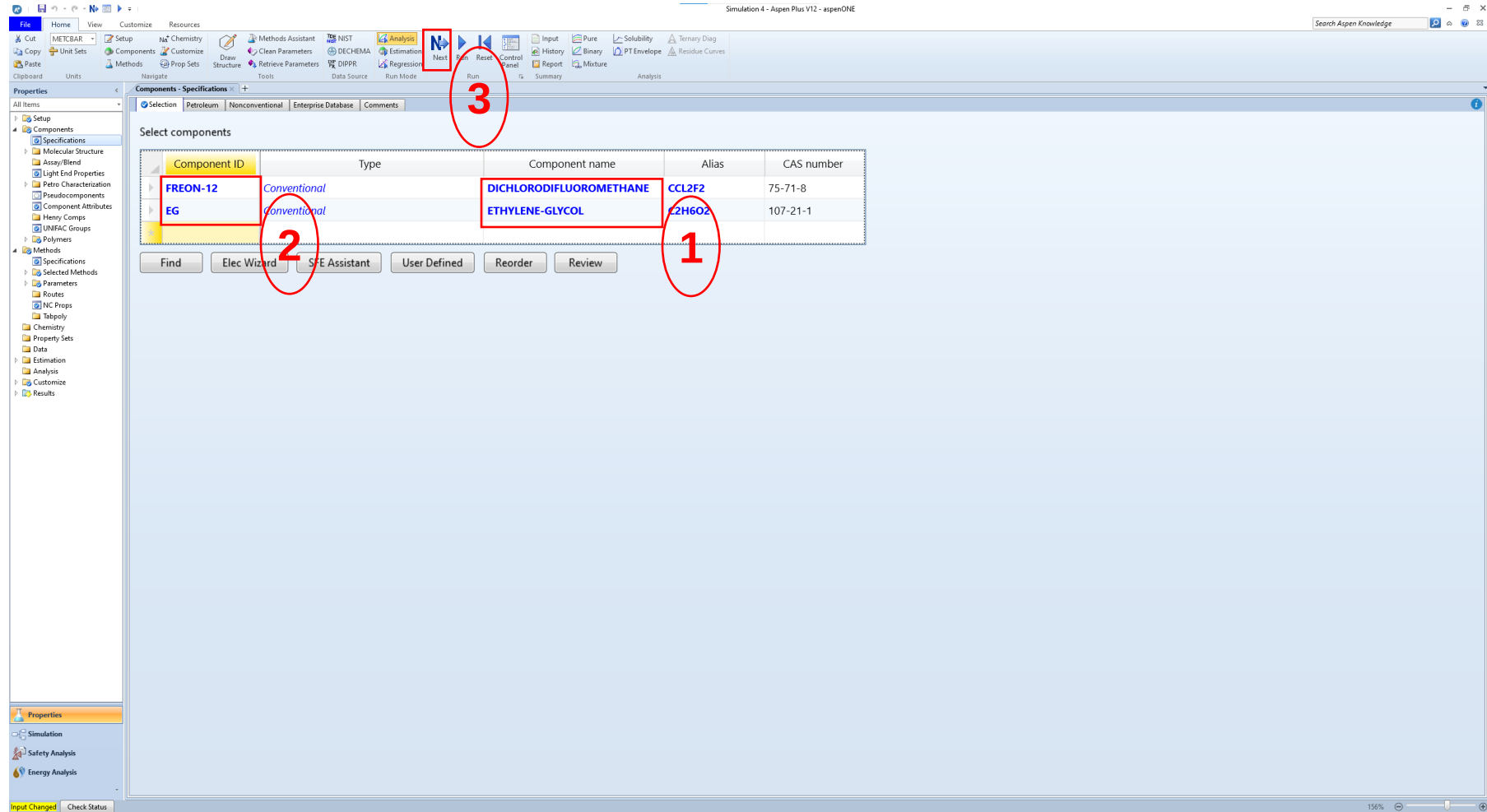
Example 2.3 Problem Analysis

- Purpose of the heat exchange process: to **heat up Freon-12 from 240 K to 300 K**
 - **Hot stream:** Ethylene Glycol
 - **Cold stream:** Freon-12
 - Property method: **NRTL-RK** (NRTL for the property of liquid and RK for the property of vapor not more than 10 atm)
 - Model: **Heater**
- Only ONE feed specification (Freon-12)** is required &
- Min temperature approach** cannot be included.

Schematic:



Create a blank simulation, list the components, rename the components and press next



Simulation 4 - Aspen Plus V12 - aspenONE

File Home View Customize Resources

Clipboard Units

Properties

All Items

Setup

Components

Specifications

Molecular Structure

Assay/Blend

Light End Properties

Petro Characterization

Pseudocomponents

Component Attributes

Henry Comps

UNIPAC Groups

Polymers

Methods

Specifications

Selected Methods

Parameters

Routes

HC Preps

Tabpoly

Chemistry

Property Sets

Data

Estimation

Analysis

Customize

Results

Components - Specifications

Selection Petroleum Nonconventional Enterprise Database Comments

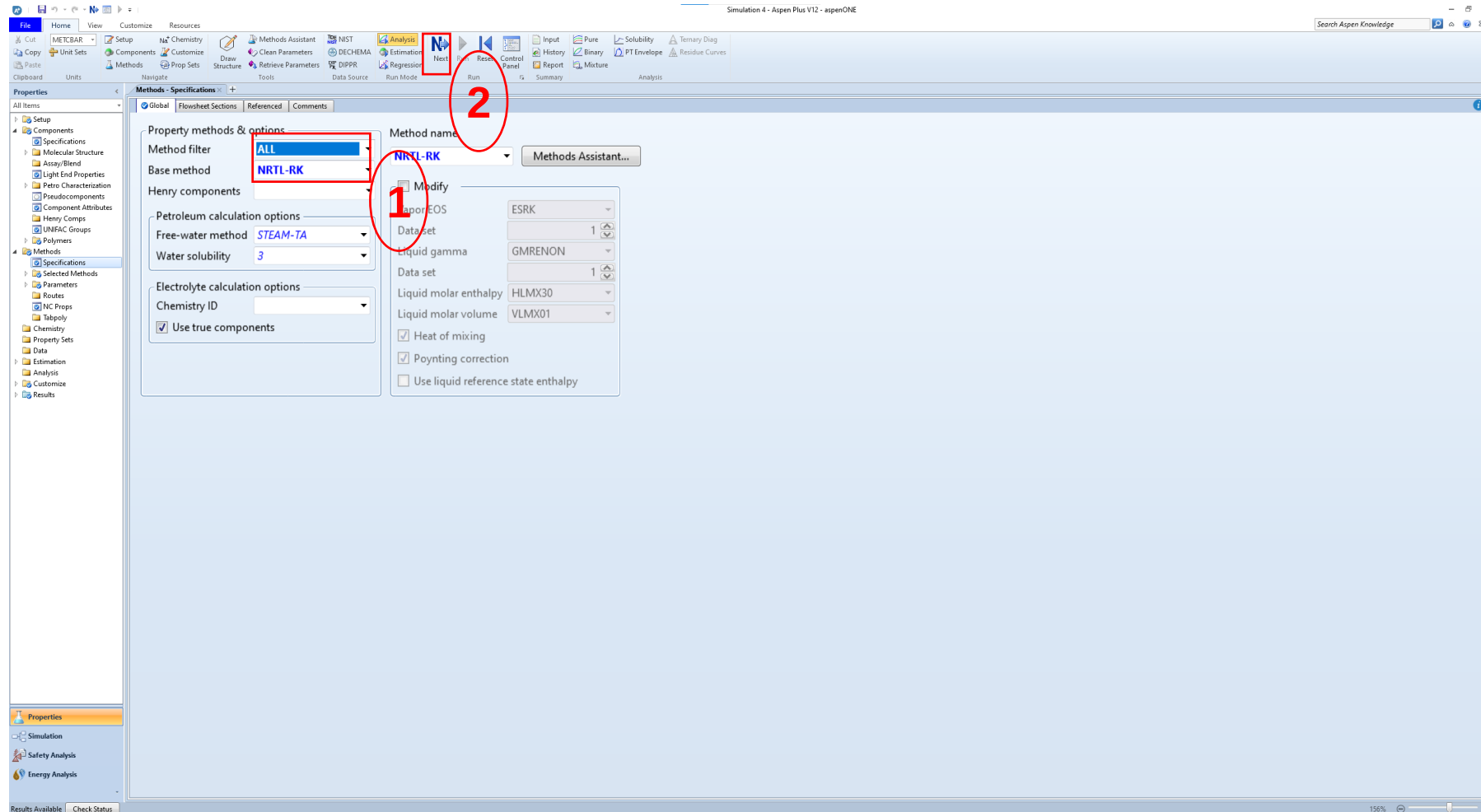
Select components

Component ID	Type	Component name	Alias	CAS number
FREON-12	Conventional	DICHLORODIFLUOROMETHANE	CCL2F2	75-71-8
EG	Conventional	ETHYLENE-GLYCOL	C2H6O2	107-21-1

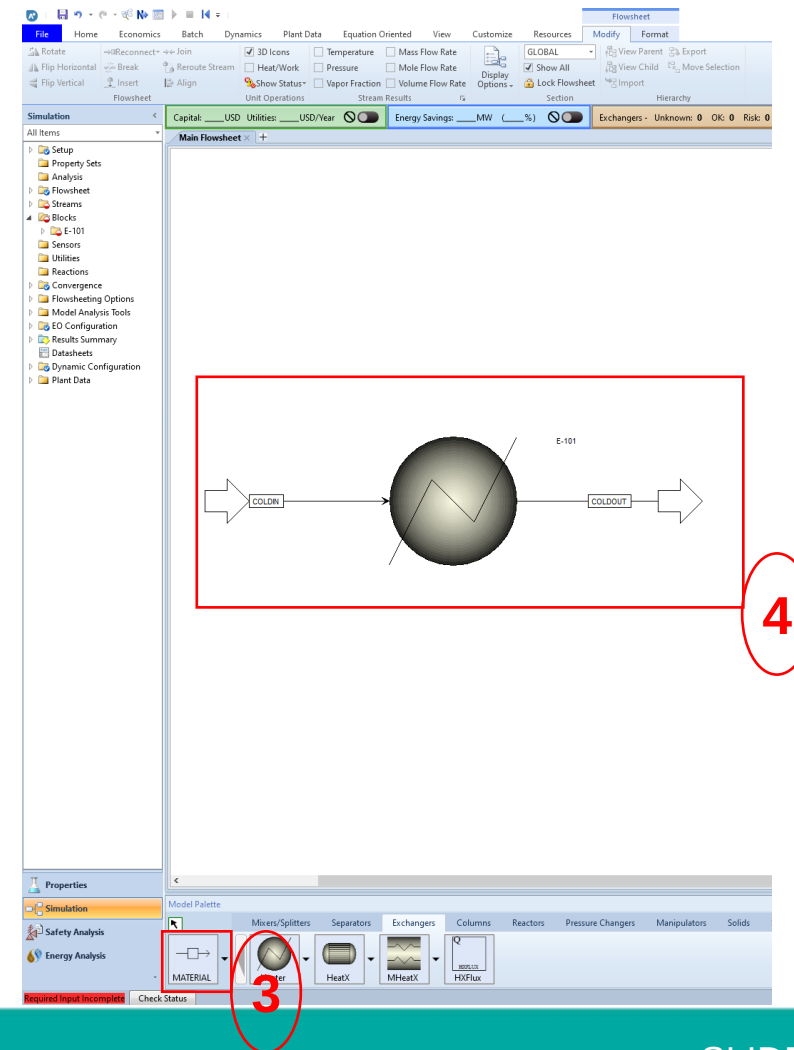
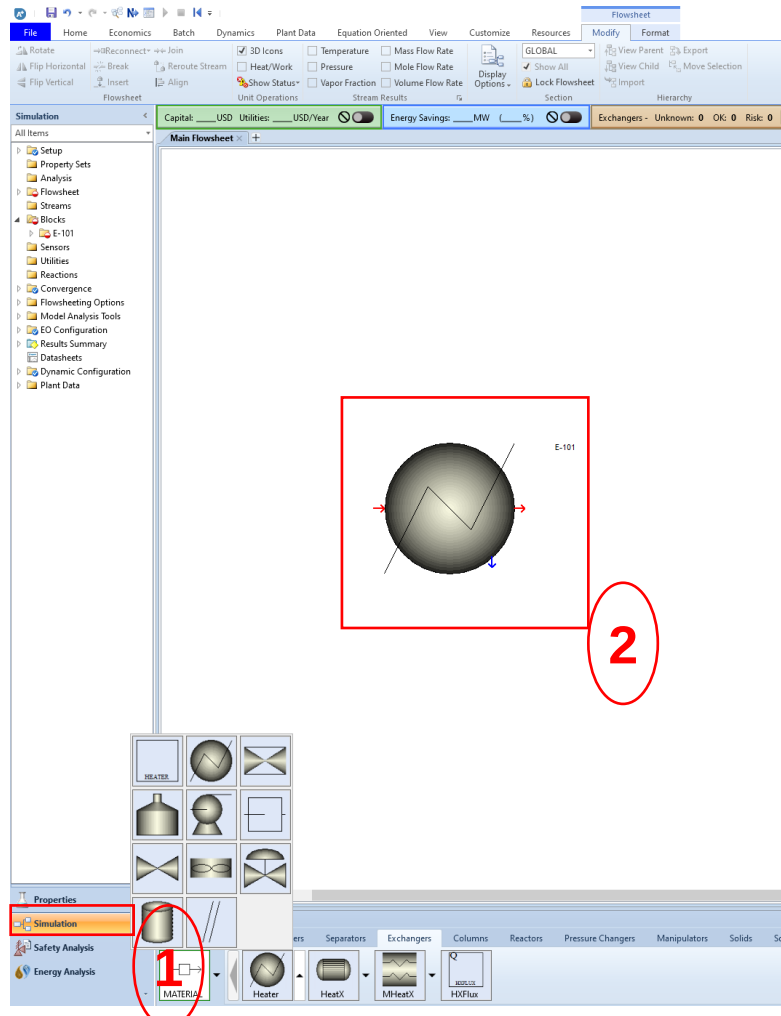
Find Elec Wizard SFE Assistant User Defined Reorder Review

Input Changed Check Status

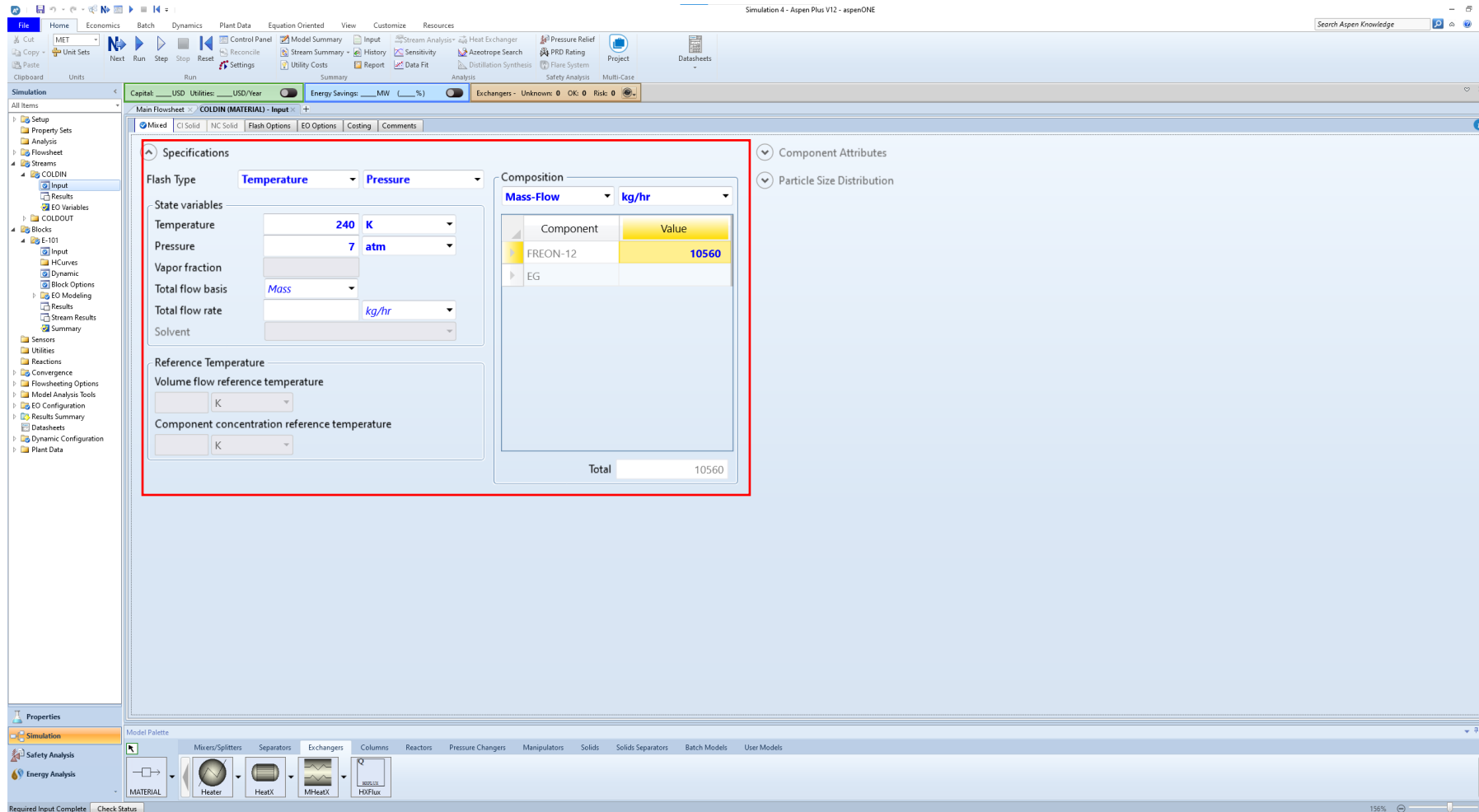
Select NRTL-RK as the property method, select next and run the simulation to generate the property sets



Shift to the simulation environment and construct the flowsheet



Input feed information



Simulation 4 - Aspen Plus V12 - aspenONE

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

Clipboard Units Run Run Stop Reset Control Panel Model Summary Input Stream Analysis Heat Exchanger Pressure Relief Project Datasheets

Simulation Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers: Unknown: 0 OK: 0 Risk: 0

Main Flowsheet COLDIN (MATERIAL) - Input

Specifications

Flash Type Temperature Pressure

State variables

Temperature 240 K

Pressure 7 atm

Vapor fraction

Total flow basis Mass

Total flow rate kg/hr

Solvent

Reference Temperature

Volume flow reference temperature K

Component concentration reference temperature K

Composition

Mass-Flow kg/hr

Component	Value
FREON-12	10560
EG	
Total	10560

Component Attributes

Particle Size Distribution

Model Palette

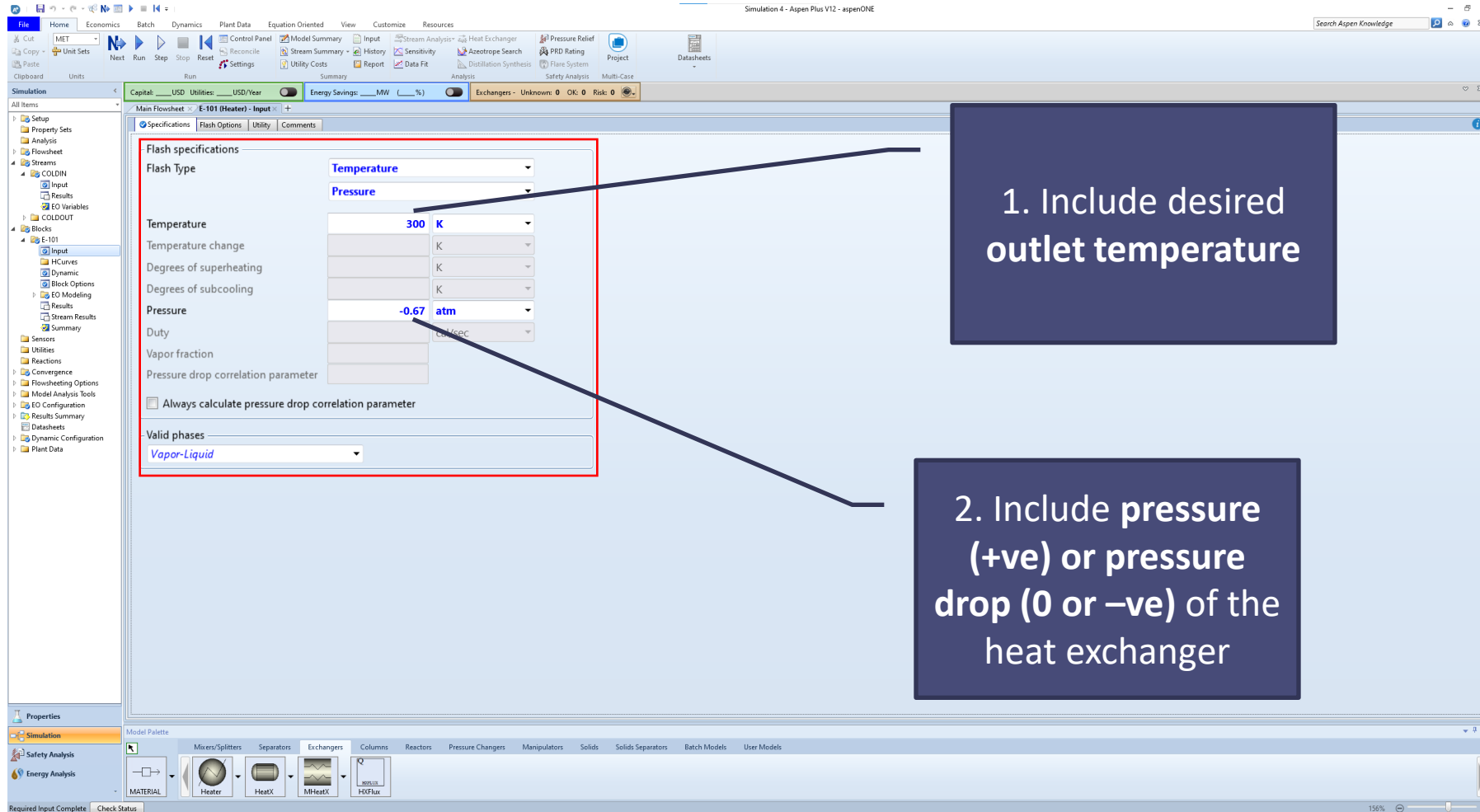
Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids Solids Separators Batch Models User Models

MATERIAL Heater HeatX MHeatX HXFlux

Required Input Complete Check Status

156%

Input heater specification information



Simulation 4 - Aspen Plus V12 - aspenONE

Flash specifications

Flash Type: Temperature, Pressure

Temperature: 300 K

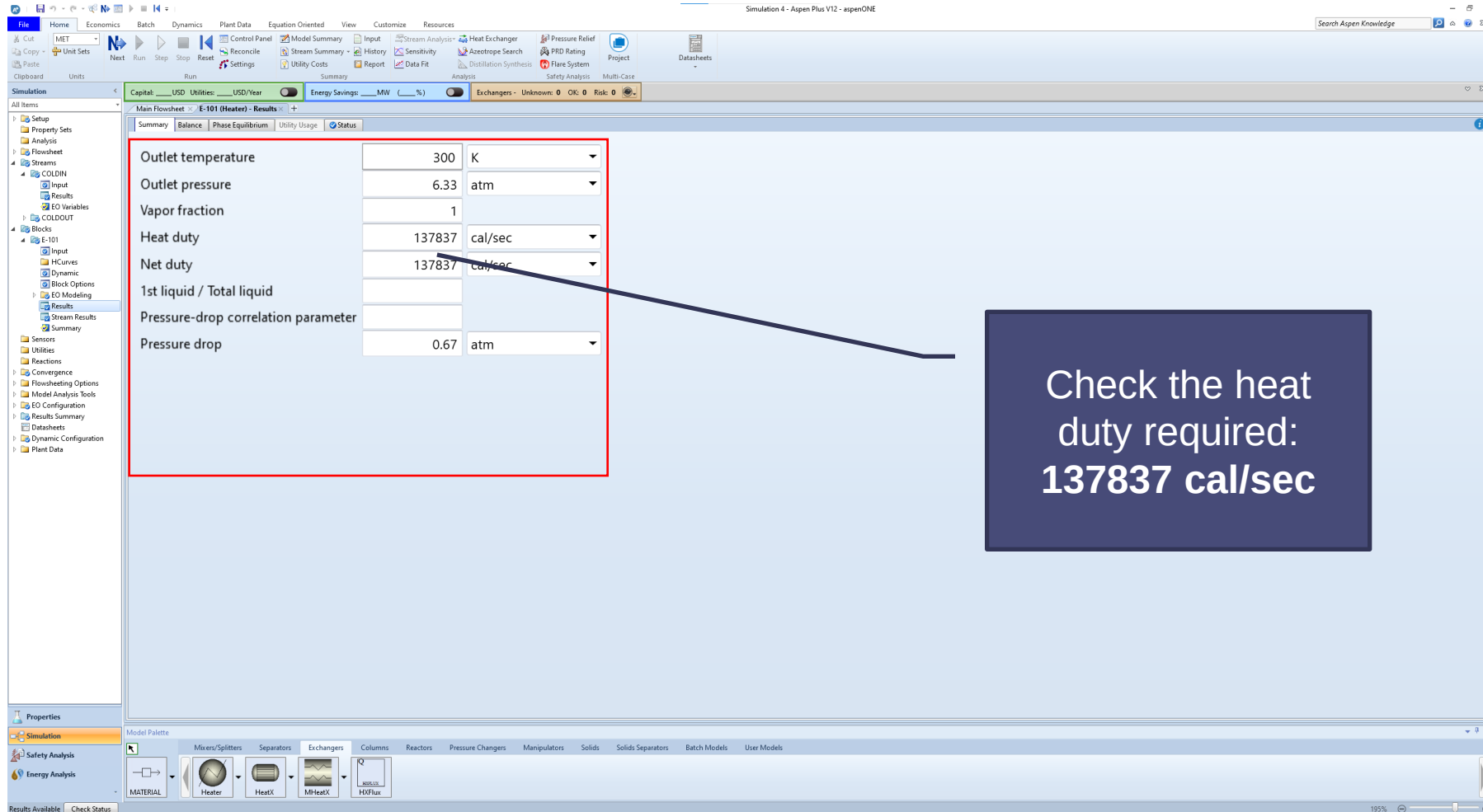
Pressure: -0.67 atm

Valid phases: Vapor-Liquid

1. Include desired outlet temperature

2. Include pressure (+ve) or pressure drop (0 or -ve) of the heat exchanger

Run the simulation and check the results folder for the heat duty required



Simulation 4 - Aspen Plus V12 - aspenONE

Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers: Unknown: 0 OK: 0 Risk: 0

Main Flowsheet > E-101 (Heater) - Results

Outlet temperature	300	K
Outlet pressure	6.33	atm
Vapor fraction	1	
Heat duty	137837	cal/sec
Net duty	137837	cal/sec
1st liquid / Total liquid		
Pressure-drop correlation parameter		
Pressure drop	0.67	atm

Check the heat duty required:
137837 cal/sec

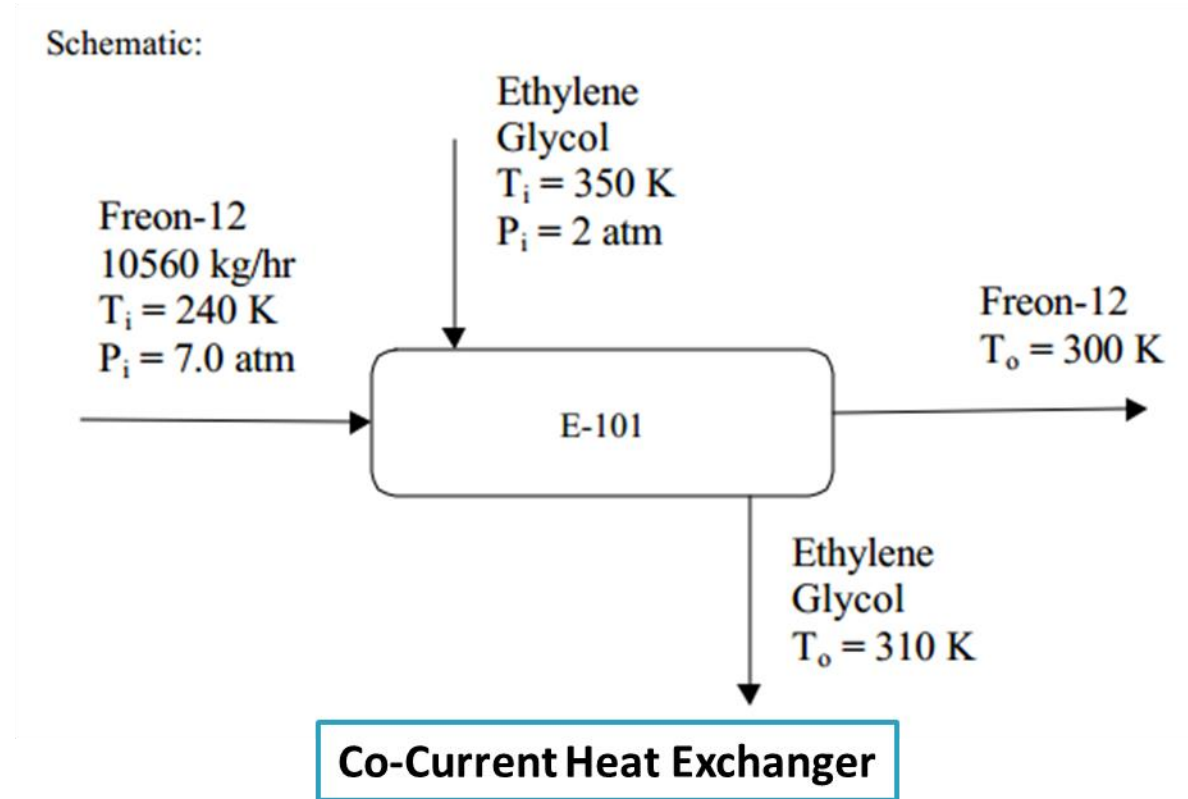
Example 2.3

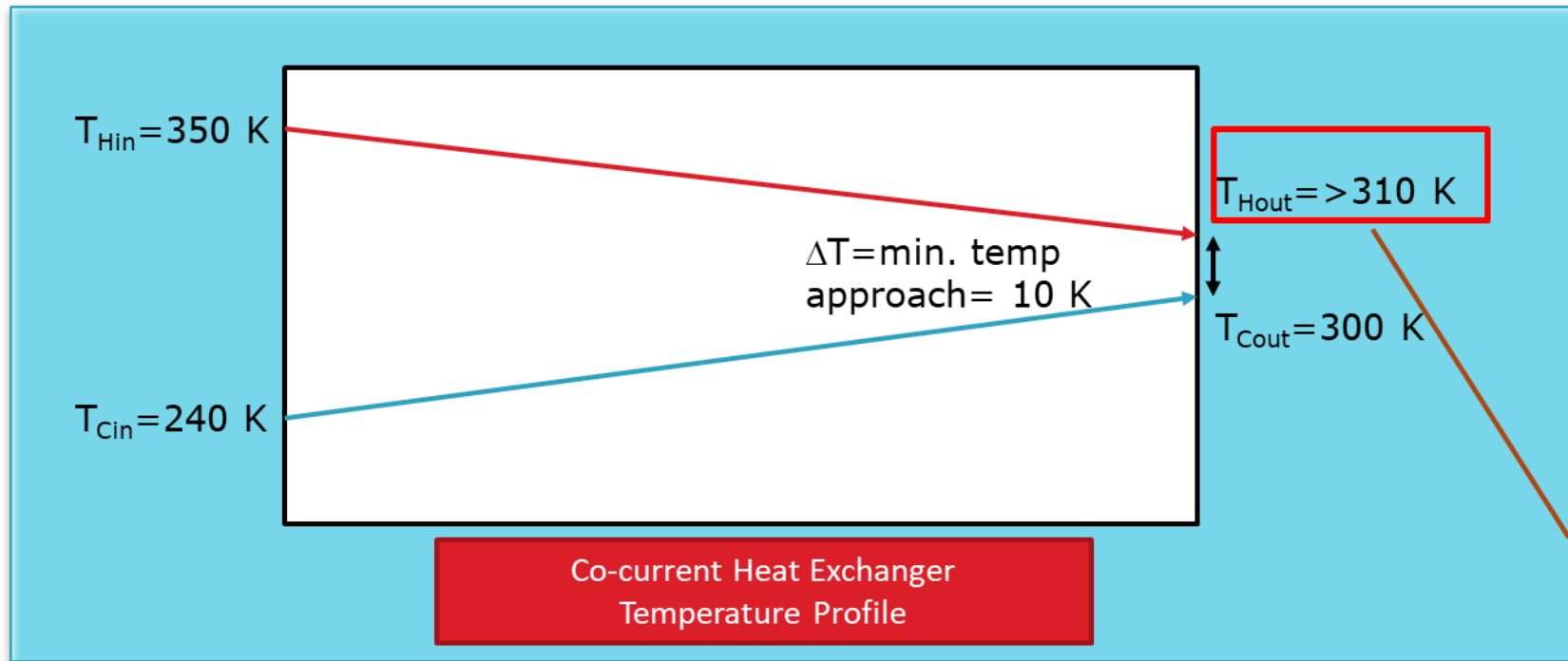
Freon-12 (Dichlorodifluoromethane), at a flow rate of 10560 kg/hr, needs to be heated from 240 K to 300 K. Ethylene glycol is available at 350 K. A typical co-current shell and tube heat exchanger will be used. The plant manager recommends that the minimum temperature approach should be at least 10 K. He also recommends using carbon-steel tubing with a pressure drop not exceeding 10 psig (0.67 atm) on **the shell side** which carries the ethylene glycol.

Simulate the heat exchange process using **HeatX model** and determine the **heat duty required** and **heat transfer area**.

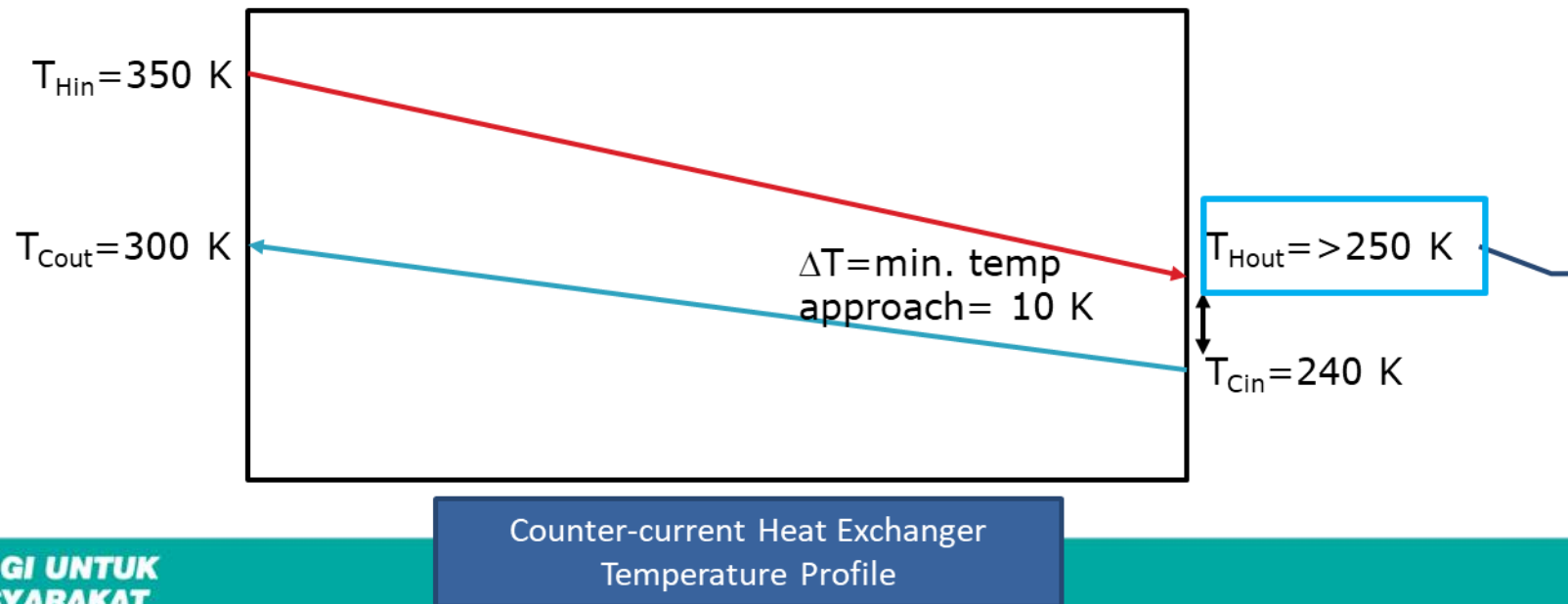
Example 2.3 Problem Analysis

- Purpose of the heat exchange process: to **heat up Freon-12 from 240 K to 300 K**
 - **Hot stream:** Ethylene Glycol (EG), 95 kmol/h
 - **Cold stream:** Freon-12
 - Property method: **NRTL-RK** (NRTL for the property of liquid and RK for the property of vapor not more than 10 atm)
 - Model: **HeatX**
- Two feeds' specifications (Freon-12 and EG)** are required &
- Min temperature approach needs** to be included.





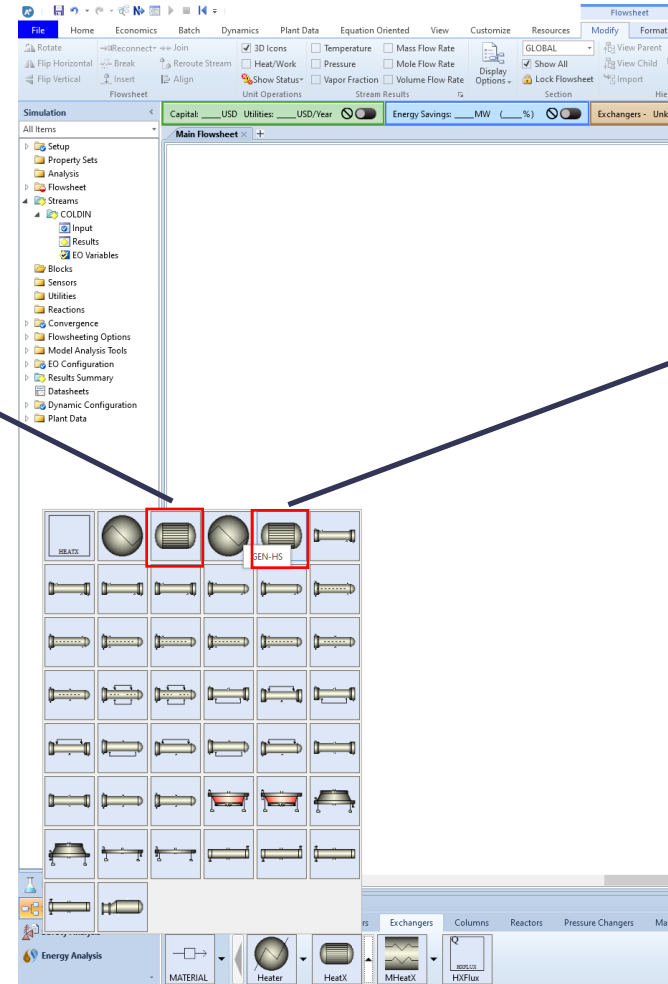
T_{out} of EG must be $> 310 \text{ K}$ to keep min. temperature approach of 10 K.



T_{out} of EG must be $> 250 \text{ K}$ to keep min. temperature approach of 10 K.

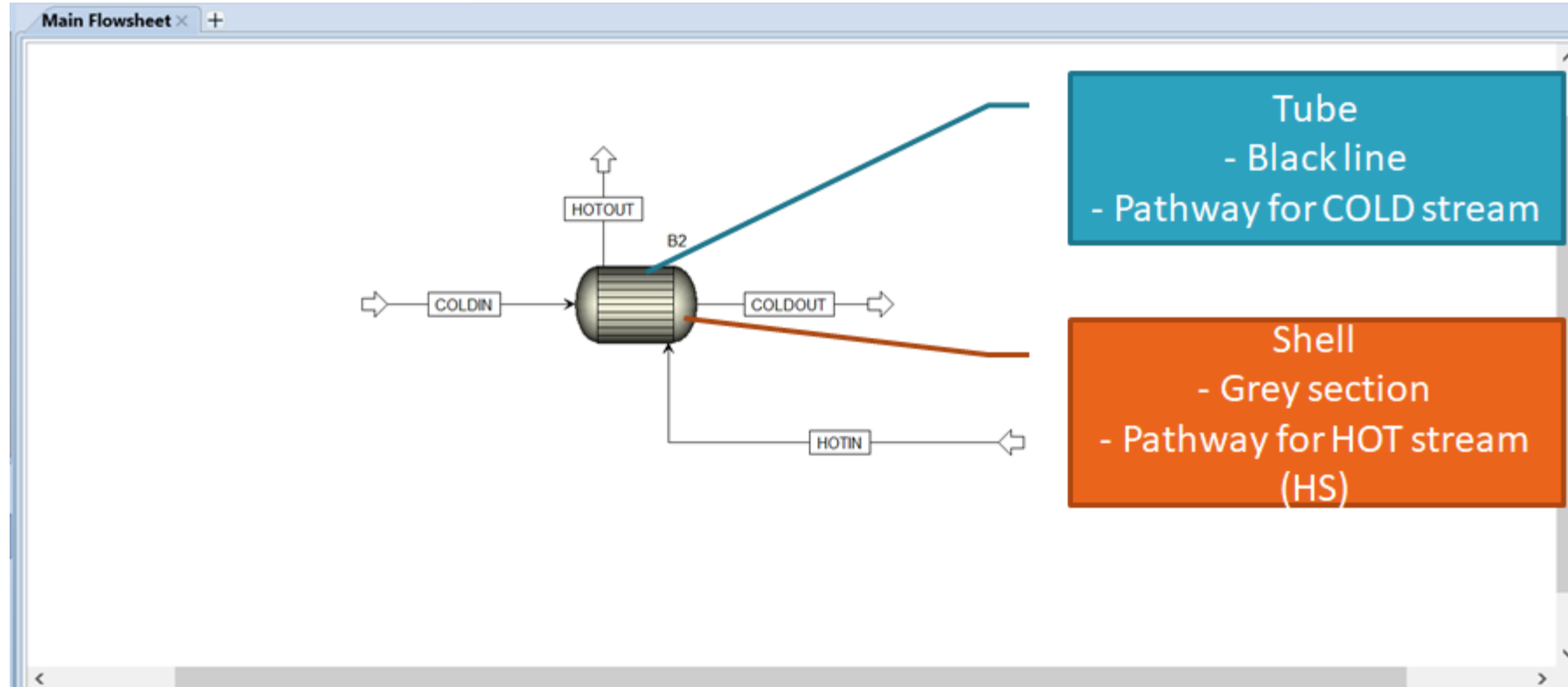
Choosing the correct HeatX model

Although visually similar when your cursor hovers over the icons, a text prompt indicates if it is HS (hot shell) or HT (hot tube). Choosing the correct icon is crucial because the ports correspond to the hot and cold streams based on the icon selected.

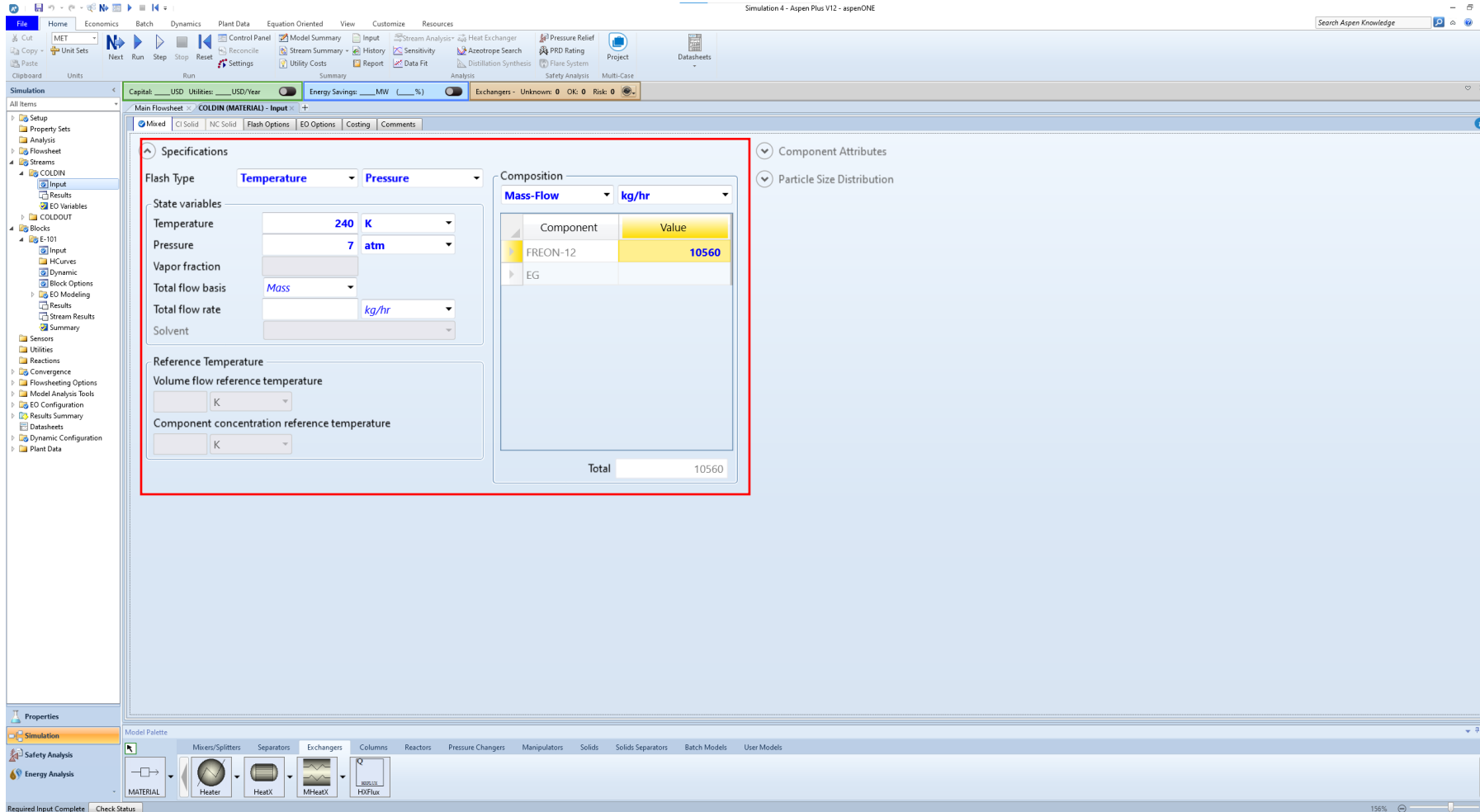


Since the problem specifies that Freon-12 (cold) is flowing through the tube the model selected must be from the Hot Shell (HS) options

When connecting the streams to a HeatX block, hovering over the ports will indicate how the streams should be oriented based on the temperature of the streams



Press next or F4 which will take you to the cold stream feed input tab



Simulation 4 - Aspen Plus V12 - aspenONE

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

Clipboard Units Run Run Settings Summary Analysis Project Data Sheets

Simulation Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers: Unknown: 0 OK: 0 Risk: 0

Main Flowsheet COLDIN (MATERIAL) - Input

Specifications

Flash Type Temperature Pressure

State variables

Temperature 240 K

Pressure 7 atm

Vapor fraction

Total flow basis Mass

Total flow rate kg/hr

Solvent

Reference Temperature

Volume flow reference temperature K

Component concentration reference temperature K

Composition

Mass-Flow kg/hr

Component	Value
FREON-12	10560
EG	

Total 10560

Component Attributes

Particle Size Distribution

Model Palette

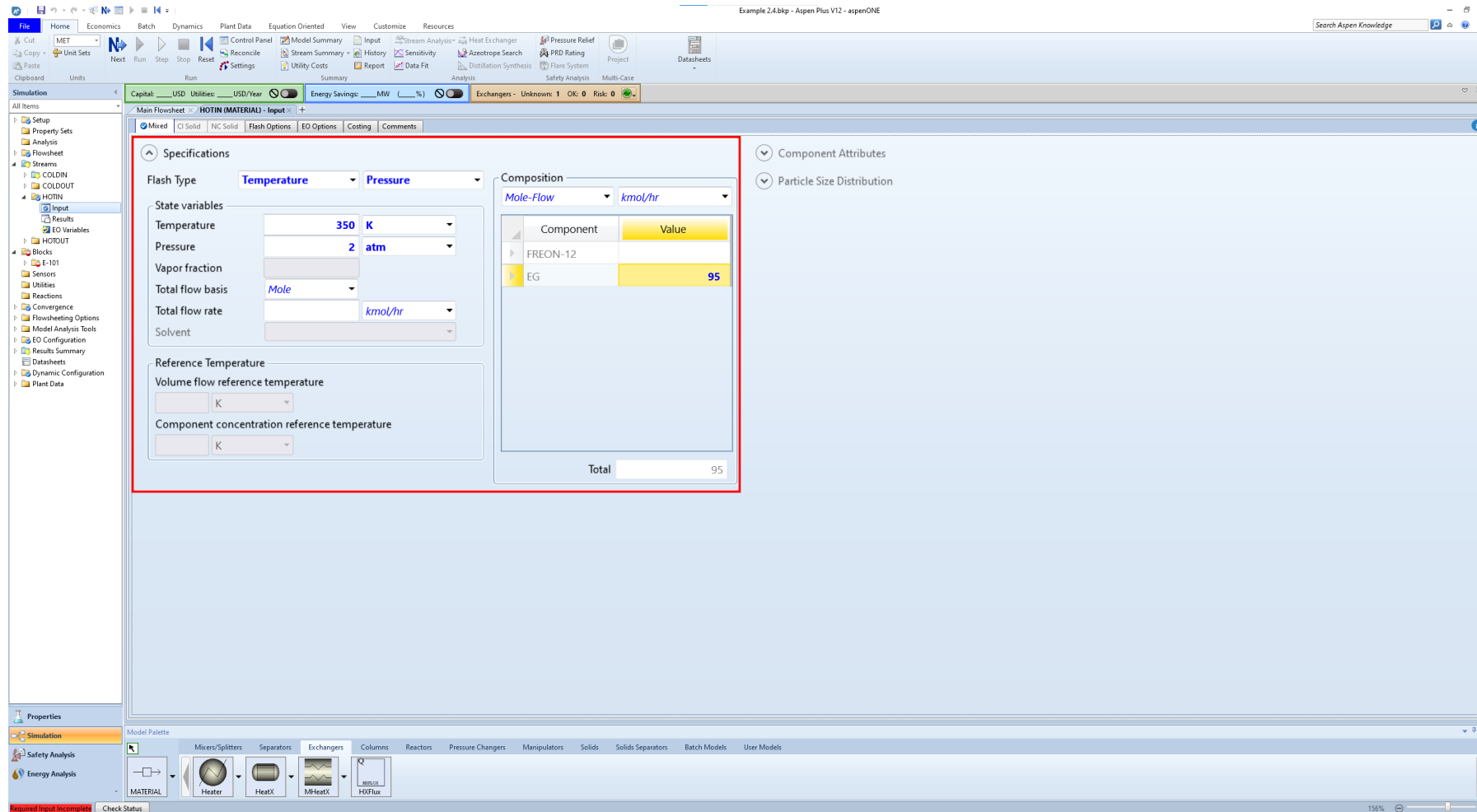
Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids Solids Separators Batch Models User Models

MATERIAL Heater HeatX MHeatX HXFlux

Required Input Complete Check Status

156%

Press next or F4 again to take you to the hot stream feed input tab



Example 2.4.bbp - Aspen Plus V12 - aspenONE

Simulation: Capital: USD/Year Utilities: USD/Year Energy Savings: MW (%) Exchangers - Unknown: 1 OK: 0 Risk: 0

Main Flowsheet > HOTIN (MATERIAL) - Input

Specifications

Flash Type: Temperature Pressure

State variables

Temperature: 350 K

Pressure: 2 atm

Vapor fraction

Total flow basis: Mole

Total flow rate: kmol/hr

Solvent

Reference Temperature

Volume flow reference temperature

Component concentration reference temperature

Composition

Mole-Flow kmol/hr

Component	Value
FREON-12	
EG	95
Total	95

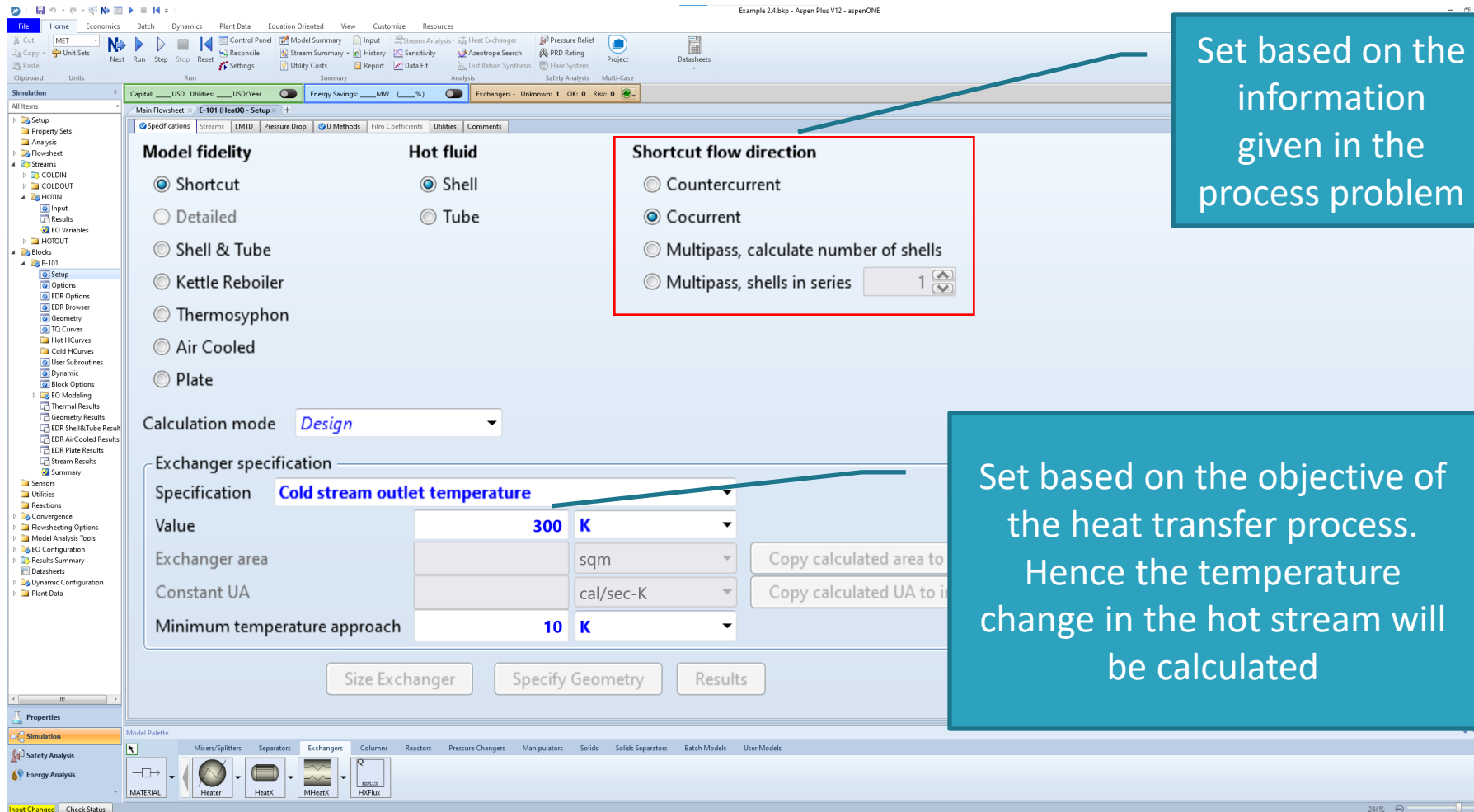
Component Attributes

Particle Size Distribution

Model Palette: Mixers/Splitters, Separators, Exchangers, Columns, Reactors, Pressure Changers, Manipulators, Solids, Solids Separators, Batch Models, User Models

Required Input Incomplete Check Status

Press next or F4 yet again will take you to the HeatX setup tab.



The screenshot shows the Aspen Plus V12 interface with the HeatX Setup tab selected. The left sidebar shows the project hierarchy, and the main window displays the HeatX configuration options. A red box highlights the 'Shortcut flow direction' section, and a blue box highlights the 'Exchanger specification' section. A blue callout box on the right explains the settings.

Set based on the information given in the process problem

Set based on the objective of the heat transfer process. Hence the temperature change in the hot stream will be calculated

Model fidelity

- ☒ Shortcut
- ☐ Detailed
- ☐ Shell & Tube
- ☐ Kettle Reboiler
- ☐ Thermosiphon
- ☐ Air Cooled
- ☐ Plate

Hot fluid

- ☒ Shell
- ☐ Tube

Shortcut flow direction

- ☐ Countercurrent
- ☒ Cocurrent
- ☐ Multipass, calculate number of shells
- ☐ Multipass, shells in series 1

Calculation mode Design

Exchanger specification

Specification: Cold stream outlet temperature

Value: 300 K

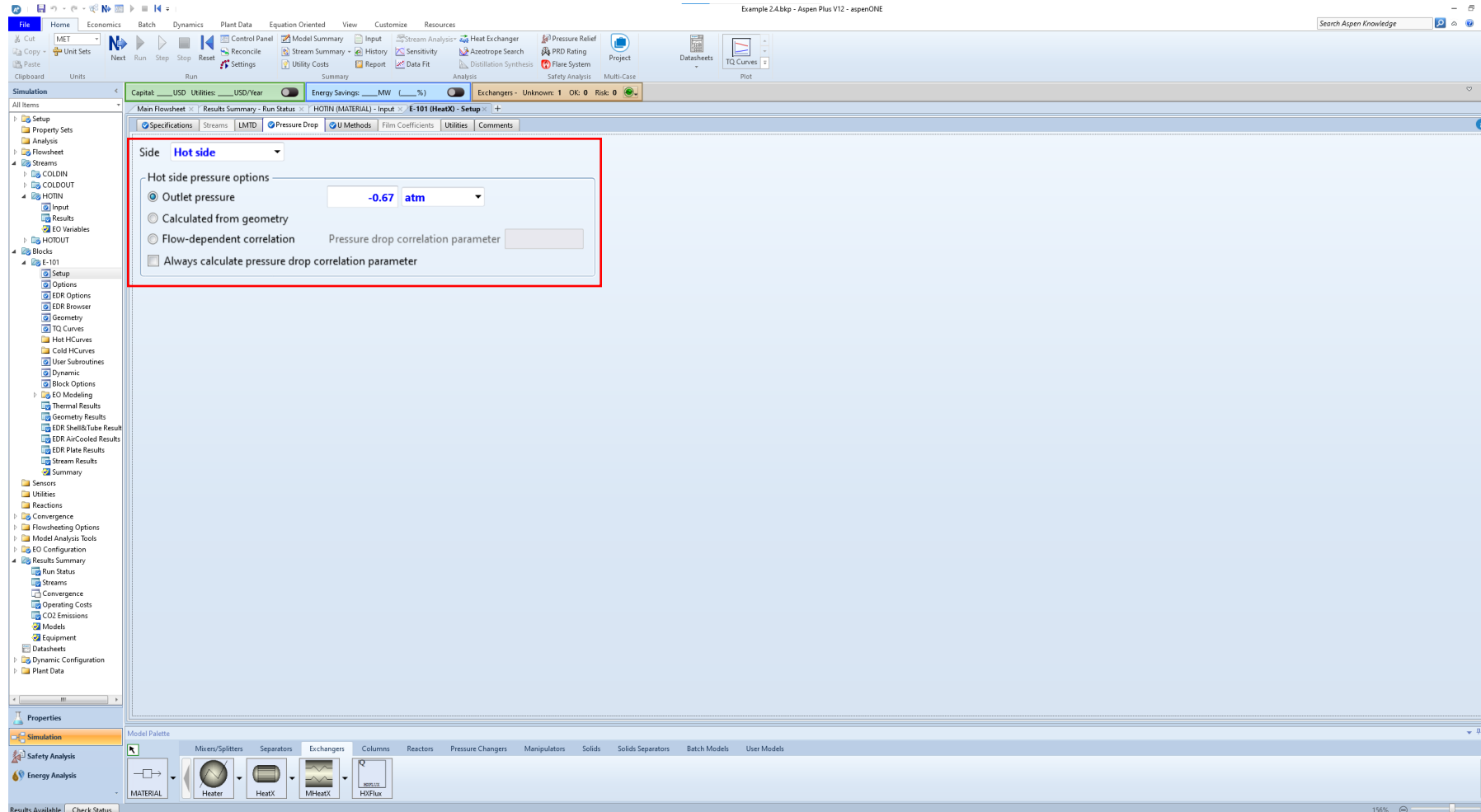
Exchanger area: sqm

Constant UA: cal/sec-K

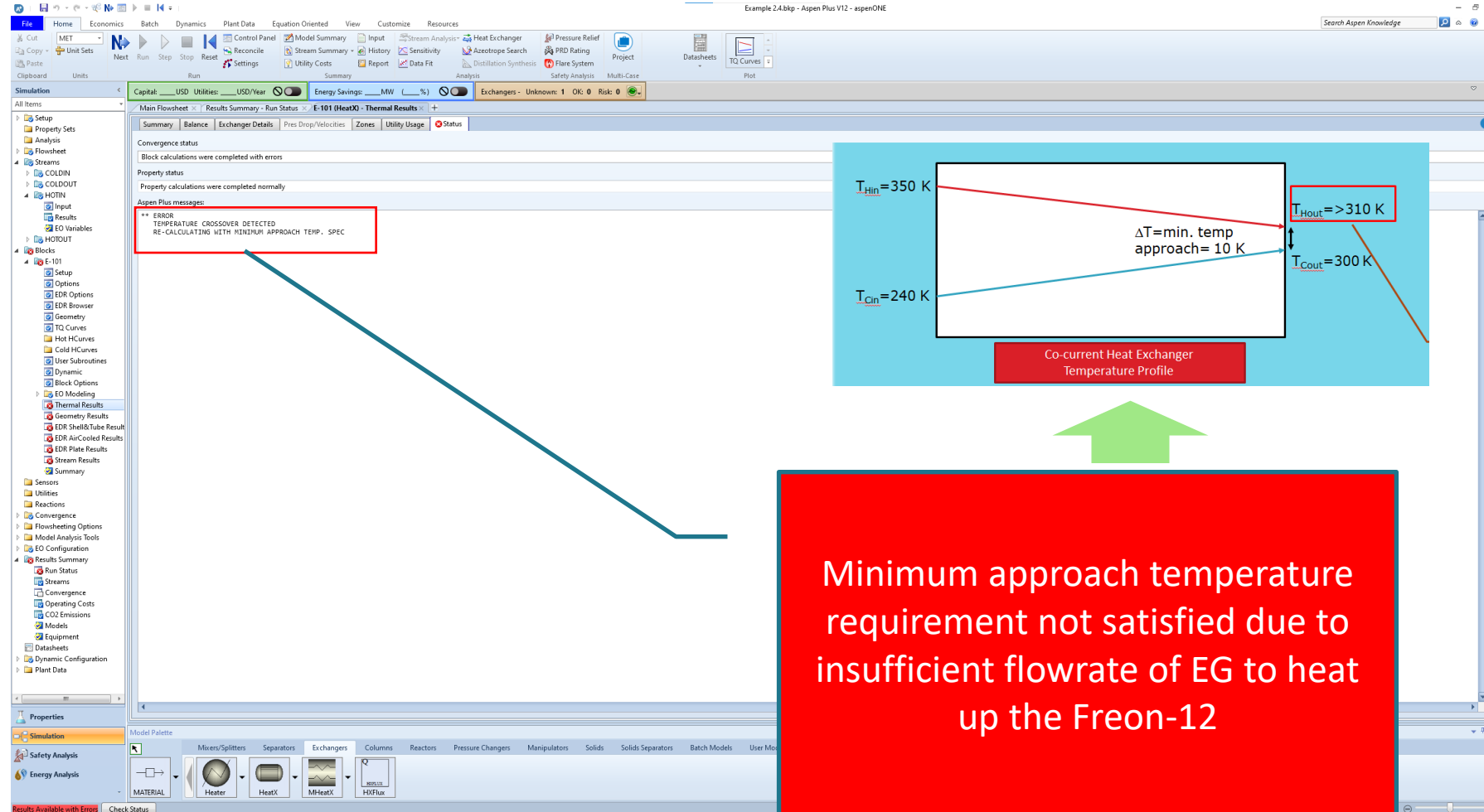
Minimum temperature approach: 10 K

Buttons: Size Exchanger, Specify Geometry, Results

Pressure drop across shell (hot) input as specified in the problem and simulation can be run as all required input is complete



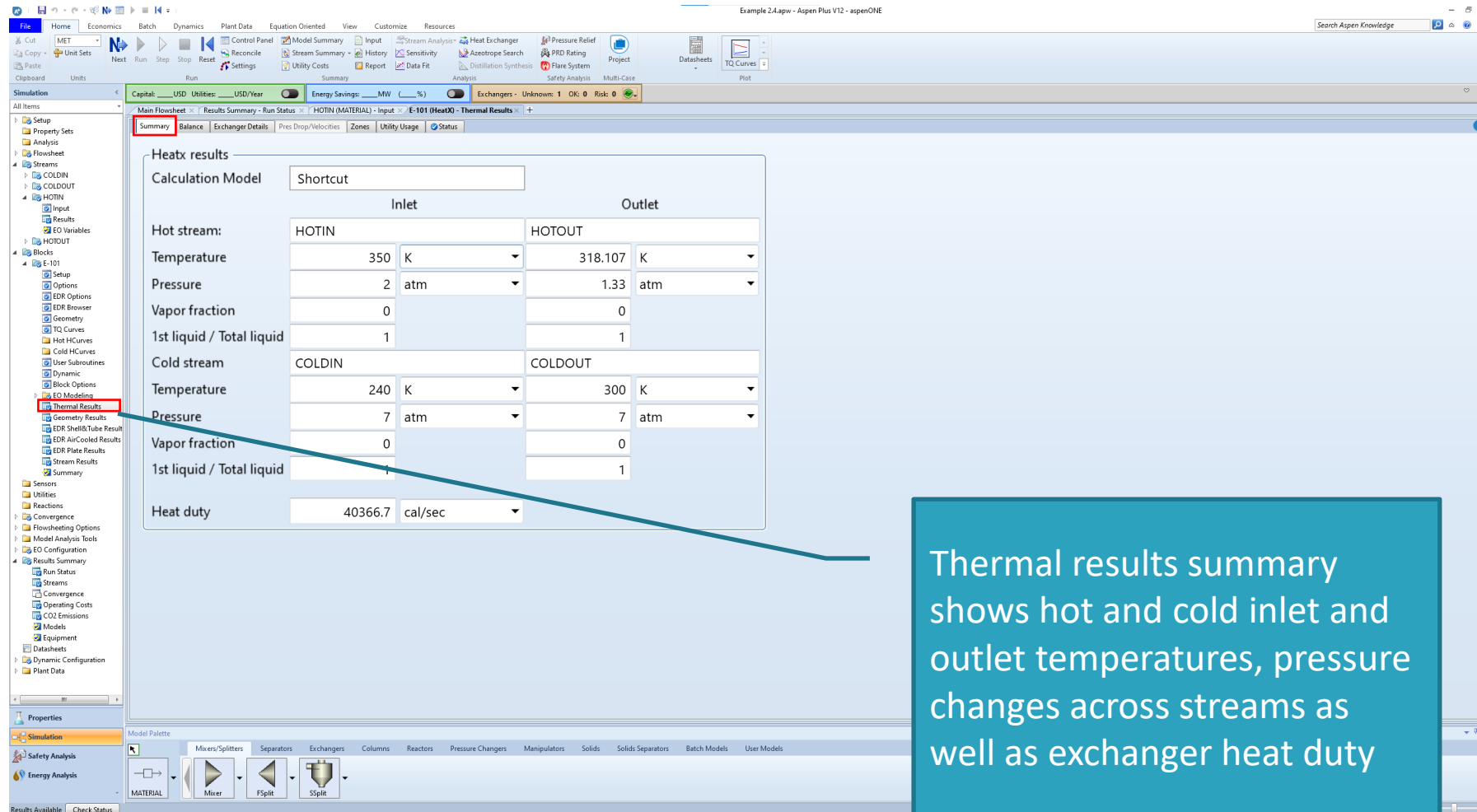
Simulation runs with errors, by clicking on the status button followed by the block that contains the error the source of error can be identified



To remedy this error, the flowrate of EG is raised to 120kmol/hr and the simulation is run once more

Simulation able to run without error because EG exit temperature is now > 310K

Most important information can be obtained from the thermal results summary



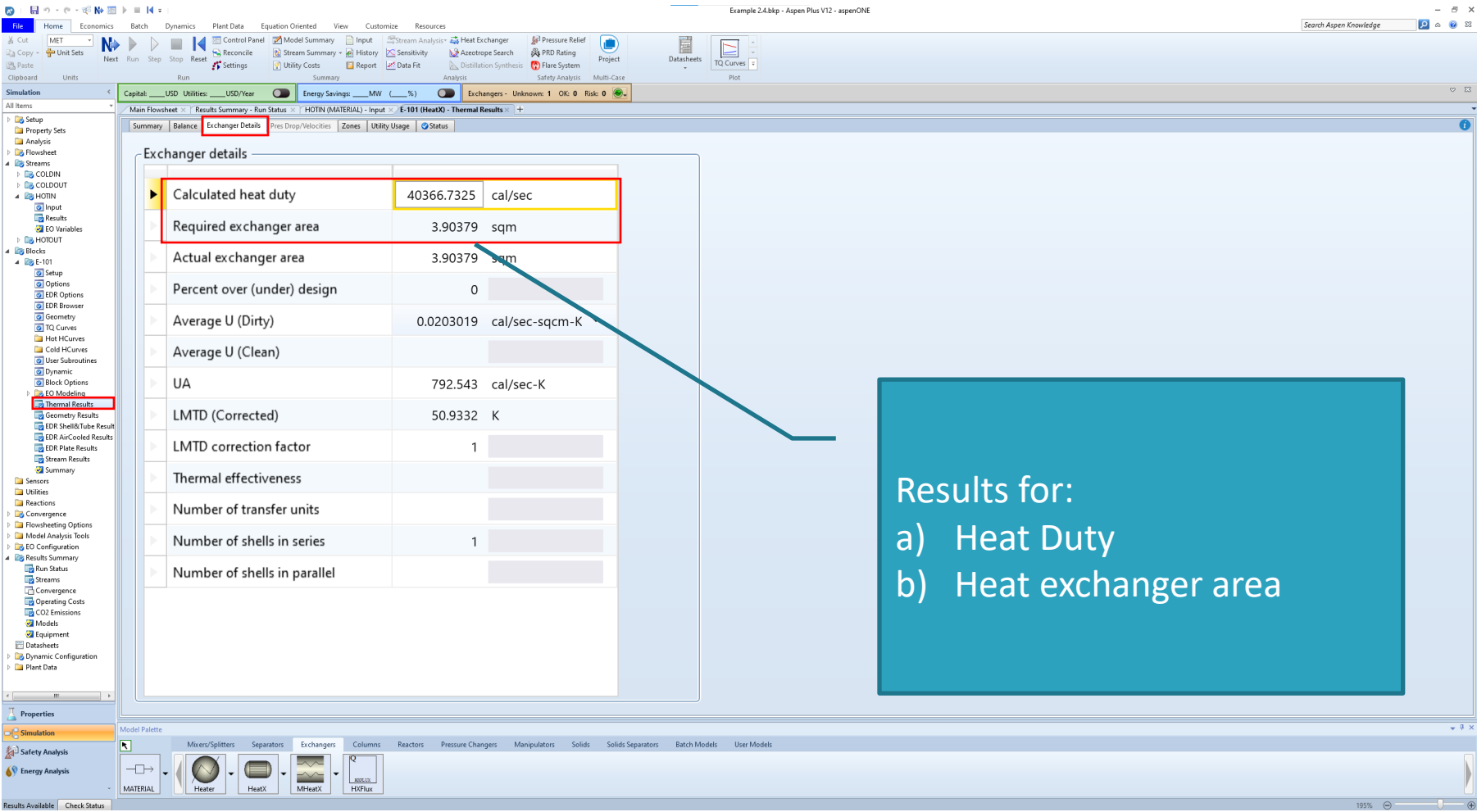
Heatx results

Calculation Model:

	Inlet		Outlet	
Hot stream:	HOTIN		HOTOUT	
Temperature	350	K	318.107	K
Pressure	2	atm	1.33	atm
Vapor fraction	0		0	
1st liquid / Total liquid	1		1	
Cold stream:	COLDIN		COLDOUT	
Temperature	240	K	300	K
Pressure	7	atm	7	atm
Vapor fraction	0		0	
1st liquid / Total liquid	1		1	
Heat duty	40366.7 cal/sec			

Thermal results summary shows hot and cold inlet and outlet temperatures, pressure changes across streams as well as exchanger heat duty

Problem solution can be found under thermal results > exchanger details tab



Example 2.4.bkp - Aspen Plus V12 - aspenONE

Search Aspen Knowledge

Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers: Unknown: 1 OK: 0 Risk: 0

Simulation: Main Flowsheet | Results Summary - Run Status | HOTIN (MATERIAL) - Input | E-101 (HeatX) - Thermal Results

Summary | Balance | **Exchanger details** | Pres Drop/Velocities | Zones | Utility Usage | Status

Exchanger details

▶ Calculated heat duty	40366.7325	cal/sec
▶ Required exchanger area	3.90379	sqm
▶ Actual exchanger area	3.90379	sqm
▶ Percent over (under) design	0	
▶ Average U (Dirty)	0.0203019	cal/sec-sqcm-K
▶ Average U (Clean)		
▶ UA	792.543	cal/sec-K
▶ LMTD (Corrected)	50.9332	K
▶ LMTD correction factor	1	
▶ Thermal effectiveness		
▶ Number of transfer units		
▶ Number of shells in series	1	
▶ Number of shells in parallel		

Thermal Results

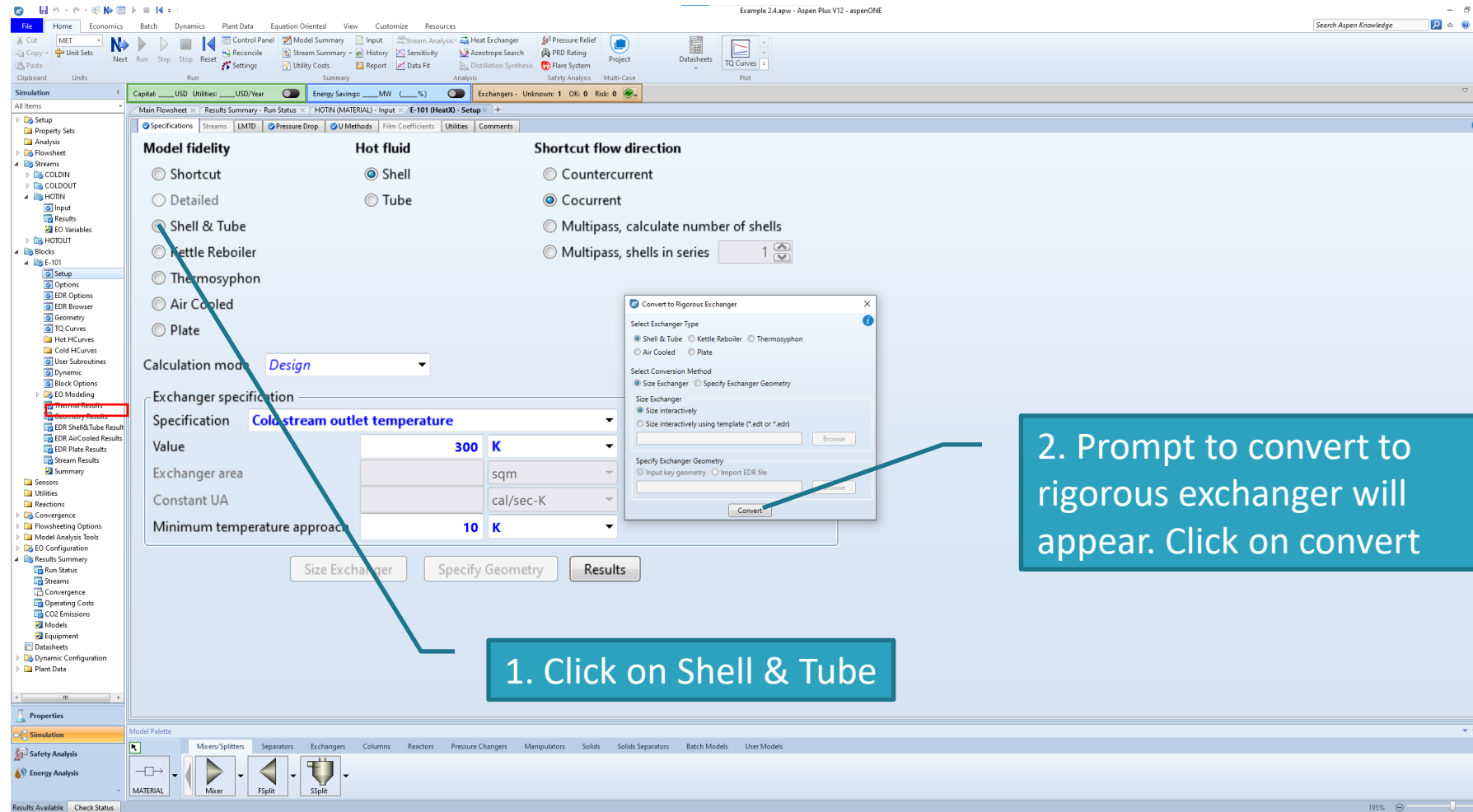
Results for:
a) Heat Duty
b) Heat exchanger area

Model Palette: Mixers/Splitters, Separators, Exchangers, Columns, Reactors, Pressure Changers, Manipulators, Solids, Solids Separators, Batch Models, User Models

Properties: Simulation, Safety Analysis, Energy Analysis

Results Available | Check Status

To perform EDR sizing you can click on any of the exchanger types to convert to a more rigorous modelling of the heat exchanger



Model fidelity

- ☐ Shortcut
- ☐ Detailed
- ☒ Shell & Tube
- ☐ Kettle Reboiler
- ☐ Thermosyphon
- ☐ Air Cooled
- ☐ Plate

Hot fluid

- ☒ Shell
- ☐ Tube

Shortcut flow direction

- ☐ Countercurrent
- ☒ Cocurrent
- ☐ Multipass, calculate number of shells
- ☐ Multipass, shells in series

Calculation mode Design

Exchanger specification

Specification: Cold stream outlet temperature

Value: 300 K

Exchanger area: sqm

Constant UA: cal/sec-K

Minimum temperature approach: 10 K

Buttons: Size Exchanger, Specify Geometry, Results

Convert to Rigorous Exchanger

Select Exchanger Type

- ☒ Shell & Tube
- ☐ Kettle Reboiler
- ☐ Thermosyphon
- ☐ Air Cooled
- ☐ Plate

Select Conversion Method

- ☒ Size Exchanger
- ☐ Specify Exchanger Geometry

Size Exchanger

- ☒ Size interactively
- ☐ Size interactively using template (*.edt or *.edr)

Specify Exchanger Geometry

- ☒ Input key geometry
- ☐ Import EDR file

Convert

2. Prompt to convert to rigorous exchanger will appear. Click on convert

1. Click on Shell & Tube

EDR Sizing Console - Size Shell&Tube (E-101)

☒ Geometry
 ☒ Process
 ☒ Errors & Warnings
 ☒ Run Status

Calculation mode: **Design (Sizing)**

Configuration
 TEMA Type: **B -** **E -** **M -**
 Tube layout option: **New (optimum) layout**
 Location of hot fluid: **Shell side**
 Tube OD / Pitch: **in** **0.75** / **0.9375**
 Tube pattern: **30-Triangular**
 Tubes are in baffle window: **Yes**
 Baffle type: **Single segmental**
 Baffle cut orientation: **Horizontal**
 Default exchanger material: **Carbon Steel** **1**

Size
 Specify some sizes for Design: **No**
 Shell ID / OD: **in** /
 Tube length: **in**
 Baffle spacing center-center (Bc): **in**
 Number of baffles:
 Number of tubes / Tube passes: /
 Shells in series:
 Shells in parallel:

Overall Results
 Excess surface (%):
 Dp-ratio Shellside / Tubeside:
 Total cost (all shells): **Dollar(US)**

Recent Previous
 Setting Plan Tube Layout

Design configuration
CAD design
Design results
Total cost results
Temperature profile

Stream Temperatures

Size Accept Design Save Cancel

✓ EDR Sizing Console - Size Shell&Tube (E-101)

☒ Geometry
 ☒ Process
 ☒ Errors & Warnings
 ☒ Run Status

	HotSide	ColdSide	Recent HotSide	Recent ColdSide	Previous HotSide	Previous ColdSide
Calculation mode: <i>Design (Sizing)</i>						
Process Conditions						
Mass flow rate	lb/h	16420.5		23280.8		
Mass flow rate multiplier		1		1		
Inlet pressure	psi	29.39		102.87		
Outlet pressure	psi	19.55		100.39		
Pressure at liquid surface in column	psi					
Inlet Temperature	°F	170.33		-27.67		
Outlet Temperature	°F					
Inlet vapor mass fraction		0		0		
Outlet vapor mass fraction						
Heat exchanged	BTU/h	576677				
Heat exchanged multiplier		1				
Process Input						
Allowable pressure drop	psi	9.85		7.25		
Fouling resistance	ft ² -h-F/BTU	0		0		
Calculated Results						
Pressure drop	psi					

Process information

Process information
from Aspen Plus

EDR Sizing Console - Size Shell&Tube (E-101)

Geometry Process Errors & Warnings Run Status

Calculation mode: Design (Sizing) Recent Previous Setting Plan Tube Layout

Configuration

TEMA Type: B - E - M -

Tube layout option: New (optimum) layout

Location of hot fluid: Shell side

Tube OD / Pitch: in 0.75 / 0.9375

Tube pattern: 30-Triangular

Tubes are in baffle window: Yes

Baffle type: Single segmental

Baffle cut orientation: Horizontal

Default exchanger material: Carbon Steel 1

Size

Specify some sizes for Design: No

Shell ID / OD: in / /

Tube length: in / /

Baffle spacing center-center (Bc): in / /

Number of baffles: / /

Number of tubes / Tube passes: / /

Shells in series: / /

Shells in parallel: / /

Overall Results

Excess surface (%): / /

Dp-ratio Shellside / Tubeside: / /

Total cost (all shells): Dollar(US)

Stream Temperatures

TS Bulk Temp. (F) + SS Bulk Temp (F)

Distance from End (in)

3. Click size

Size Accept Design Save Cancel

EDR Sizing Console - Size Shell&Tube (E-101)

☒ Geometry
 ☒ Process
 ☒ Errors & Warnings
 ☒ Run Status

Calculation mode: **Design (Sizing)**
 Recent Previous
 ☒ Setting Plan
 ☐ Tube Layout

Configuration

TEMA Type: **B -** **E -** **M -**
 BEM

Tube layout option: **New (optimum) layout**

Location of hot fluid: **Shell side**

Tube OD / Pitch: **in** **0.75** / **0.9375**
 0.75 / 0.9375 /

Tube pattern: **30-Triangular**
 30

Tubes are in baffle window: **Yes**
 Yes

Baffle type: **Single segmental**

Baffle cut orientation: **Horizontal**

Default exchanger material: **Carbon Steel** **1**
 Carbon Steel

Size

Specify some sizes for Design: **No**

Shell ID / OD: **in** / /
 8.071 / 8.625 /

Tube length: **in** / /
 64.9606 / /

Baffle spacing center-center (Bc): **in** / /
 2.1654 / /

Number of baffles: / /
 26 / /

Number of tubes / Tube passes: / /
 51 / 1 /

Shells in series: / /
 1 / /

Shells in parallel: / /
 1 / /

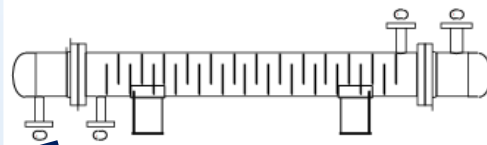
Overall Results

Excess surface (%): / /
 4 / /

Dp-ratio Shellside / Tubeside: / /
 0.2357 / 0.0865 /

Total cost (all shells): **Dollar(US)**
 10640

CAD design

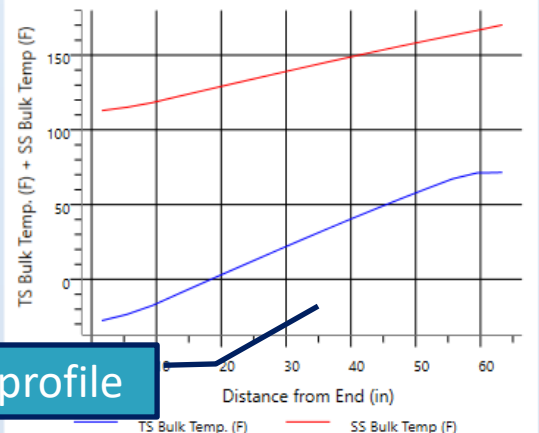


Sizing results

Total cost results

Temperature profile

Stream Temperatures



TS Bulk Temp. (F) / SS Bulk Temp. (F)

Distance from End (in)

Size Accept Design Save Cancel

EDR Sizing Console - Size Shell&Tube (E-101)

☒ Geometry
 ☒ Process
 ☒ Errors & Warnings
 ☒ Run Status

Calculation mode: **Design (Sizing)** Recent Previous ☒ Setting Plan ☐ Tube Layout

Configuration

TEMA Type: **B -** **E -** **M -** BEM

Tube layout option: **New (optimum) layout**

Location of hot fluid: **Shell side**

Tube OD / Pitch: **in** **0.75** / **0.9375** 0.75 / 0.9375

Tube pattern: **30-Triangular** 30

Tubes are in baffle window: **Yes** Yes

Baffle type: **Single segmental** Single segmental

Baffle cut orientation: **Horizontal** H

Default exchanger material: **Carbon Steel** 1 Carbon Steel

Size

Specify some sizes for Design: **No** No

Shell ID / OD: **in** 8.071 / 8.625

Tube length: **in** 64.9606

Baffle spacing center-center (Bc): **in** 2.1654

Number of baffles: **26**

Number of tubes / Tube passes: **51** / **1**

Shells in series: **1**

Shells in parallel: **1**

Overall Results

Excess surface (%): **4**

Dp-ratio Shellside / Tubeside: **0.2357** / **0.0865**

Total cost (all shells): **Dollar(US)** 10540

Stream Temperatures

4. Accept design
when satisfied

5. Save .edr file to
folder

Example 2.4.apw - Aspen Plus V12 - aspenONE

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

Simulation Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers: Unknown: 0 OK: 0 Risk: 1

Model fidelity

- Shortcut
- Detailed
- Shell & Tube**
- Kettle Reboiler
- Thermosyphon
- Air Cooled
- Plate

Hot fluid

- Shell**
- Tube

Shortcut flow direction

- Countercurrent
- Cocurrent**
- Multipass, calculate number of shells
- Multipass, shells in series 1

Calculation mode **Simulation**

Exchanger specification

Specification	Cold stream outlet temperature	
Value	300	K
Exchanger area		sqm
Constant UA		cal/sec-K
Minimum temperature approach	10	K

Size Exchanger Specify Geometry Results

Results available indicates that model has converged

Properties Simulation Safety Analysis Energy Analysis

Model Palette

MATERIAL Mixer FSplit SSplit

Example 2.4.apw - Aspen Plus V12 - aspenONE

Search Aspen Knowledge

File Home Economics Batch Dynamics Plant Data Equation Oriented View Customize Resources

Simulation Capital: USD Utilities: USD/Year Energy Savings: MW (%) Exchangers - Unknowns: 0 OK: 0 Risk: 1

Overall Thermal Delta P Setting Plan Tubesheet Layout Profile Plots

Overall results

Hot stream location	Shell	
Heat duty	39252.1	cal/sec
Required exchanger area	4.80899	sqm
Actual exchanger area	4.80097	sqm
% excess surface area	-0.166689	
Avg. heat transfer coefficient	0.0126624	cal/sec-sqcm-K
UA	608.932	cal/sec-K
LMTD (Corrected)	64.4606	K
LMTD correction factor	1.00098	
Vibration indication	NO	
High Rhov2 indication	YES	
Calculation method used by EDR	ADVANCED	

View EDR results

Model Palette: Mixers/Splitters Separators Exchangers Columns Reactors Pressure Changers Manipulators Solids Solids Separators Batch Models User Models

Properties Simulation Safety Analysis Energy Analysis

Results Available Check Status

Exercise 2.2

- Using the process information from Example 2.3 determine the minimum flowrate of ethylene glycol required to achieve the process objective successfully. You may use sensitivity analysis to help you.